

Emergent Quantum Matrix Geometry: Matrix first or State first ?



Discrete Approaches to the Dynamics of Fields
and Space-Time 2026@ Riken Kobe,
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[Refs.] arXiv 2604.24112 (to appear in PRD) and references therein.

Geometry from Quantum State

1

Standard scheme

$$\{x, y\} \longrightarrow [\hat{x}, \hat{y}]$$

Symplectic manifold

Quantization of symplectic manifold

This is the basic idea behind the conventional quantization methods:

Weyl quant., deformation quant., geometric quant., BT quant. ..
(*except for Connes' NCG*)

Restricted to symplectic manifolds $\rightarrow$ General manifold ?

Restricted to the commutator formalism $\rightarrow$ General NCG such as Nambu bracket ?

Classical Geometry $\longrightarrow$ Quantum Geometry

Present scheme

Quantum State $\longrightarrow$ Matrix Geometry

(*Connes' NCG: Algebra $\rightarrow$ Geometry*)



13

非可換微分幾何学の基礎

前田 吉昭・佐古 彰史

新井 仁之・小林 俊行・斎藤 毅・吉田 陽広 編

共立出版

Quantum mechanics and M(atrix) theory 2

This year is the centennial of wave mechanics!

Matrix mechanics (1925)

Matrix



(from wiki)



Wave mechanics (1926)

Quantum State, Operator



(from wiki)

$$F^H = \int \rho(x) u_H(x) [F, u_H(x)] dx,$$

Schroedinger (Annalen der Physik 1926)

$$O_{ij} = \langle \psi_i | O(x, p = -i\partial_x) | \psi_j \rangle$$

Matrix first or state first?

M(atrix) theory in string theory

IKKT('97) BFSS('97) . . .

$$S = \frac{1}{4} \text{tr}([A_\mu, A_\nu]^2) + \dots$$

(IIB or not IIB ?)



サイエンス社



Matrix geometry from Quantum states

3

Matrix model

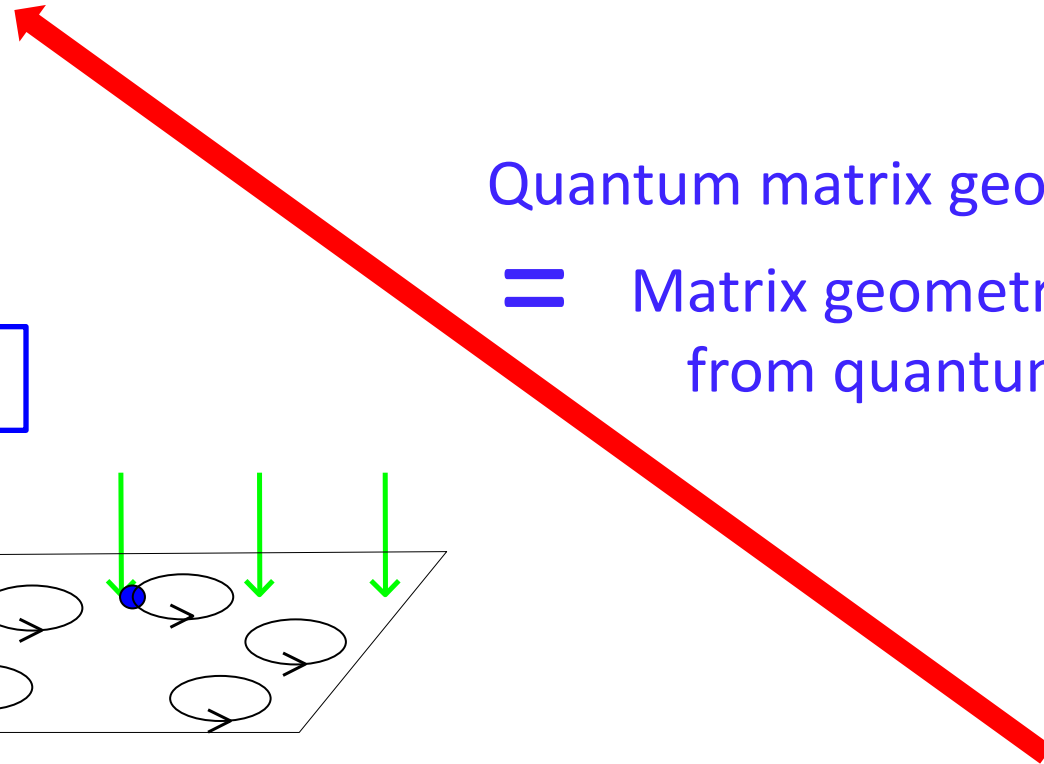
(Matrix) Geometry
(classical sol.)



Fluctuation
(field theory)



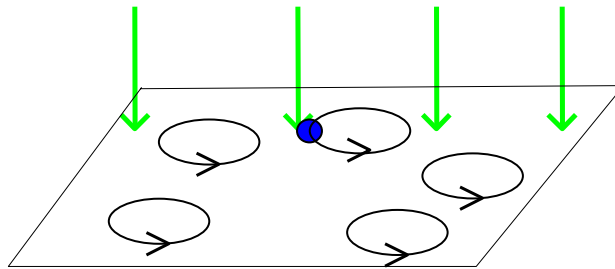
Quantum states



Quantum matrix geometry

= Matrix geometry
from quantum states

Landau model



Level projection



Quantum states



(Matrix) Geometry

My Past Studies

4

'02~ LLL geometry (higher D, SUSY ...) ``First floor''

Level projection for matrix geo.

➡ ``Toy model'' of Matrix model geometry

Explore Landau models to generate a new NCG !

'16: 2D higher LLs & fuzzy S2 ``Upper floors''

'17: fuzzy S3 & higher LLs

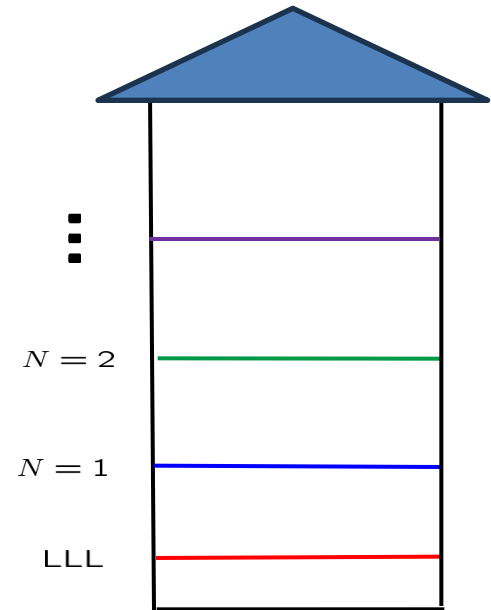
'20: fuzzy S4 & higher LLs

➡ Nested Nambu geometry

'23: Prescription for general G/H

'25: Deformed fuzzy manifold

'26: Howe duality



Ishiki's group ('18~)

Tsuchiya's group ('20~)



``Relation among all floors''

Roger Howe (from wiki)

5

Emergent Quantum Matrix Geometry I ⁵

Higher D fuzzy hyperboloid

Lowest Landau level on 4D hyperboloid

➡ 4D fuzzy hyperboloid ($\mathbb{C}P_{\text{fuzzy}}^{1,2}$)
(KH '12)

Indefinite signature higher D. fuzzy manifold

$$[X^a, X^b] = iX^{ab} \quad [X^a, X^{bc}] = i\eta^{ab}X^c - i\eta^{ac}X^b \quad \dots$$

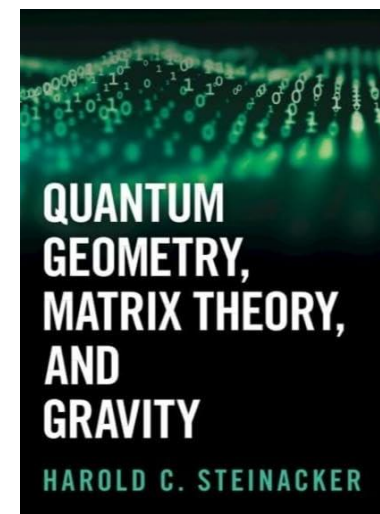
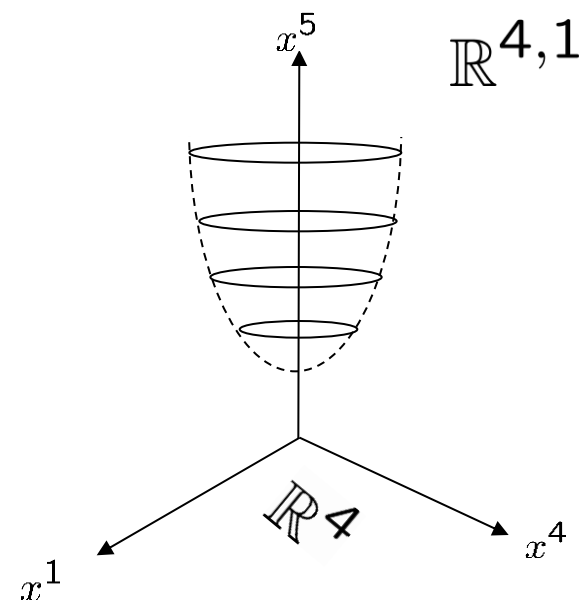
Classical sol. of indefinite signature Matrix model
with mass term

$$\eta_{bc}[X^b, [X^c, X^a]] = M^2 X^a$$

➡ Emergent gravity from IKKT (*Steinacker '18~*)
(FRW cosmology)

<https://www.youtube.com/watch?v=wQCy67ihyOk>

Chap.5 in



Emergent Quantum Matrix Geometry II

6

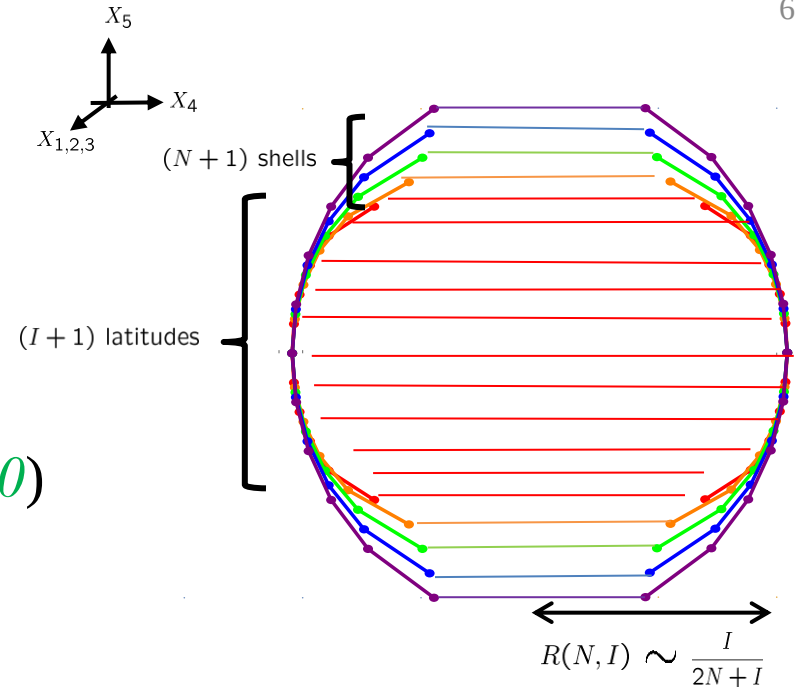
Quantum Nambu geometry

Higher Landau Level projection

➡ nested 4D fuzzy sphere
(*KH* '23, '20)

Quantum Nambu structure

$$[X_a, X_b, X_c, X_d] = 4! \epsilon_{abcde} X_e$$



➡ New solution of Yang-Mills matrix model with Chern-Simons term

$$[[X_a, X_b], X_b] = \epsilon_{abcde} X_b X_c X_d X_e$$

Emergent Quantum Matrix Geometry III

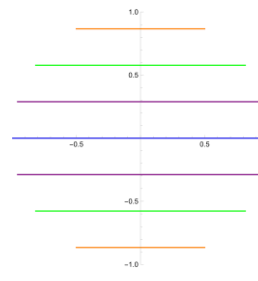
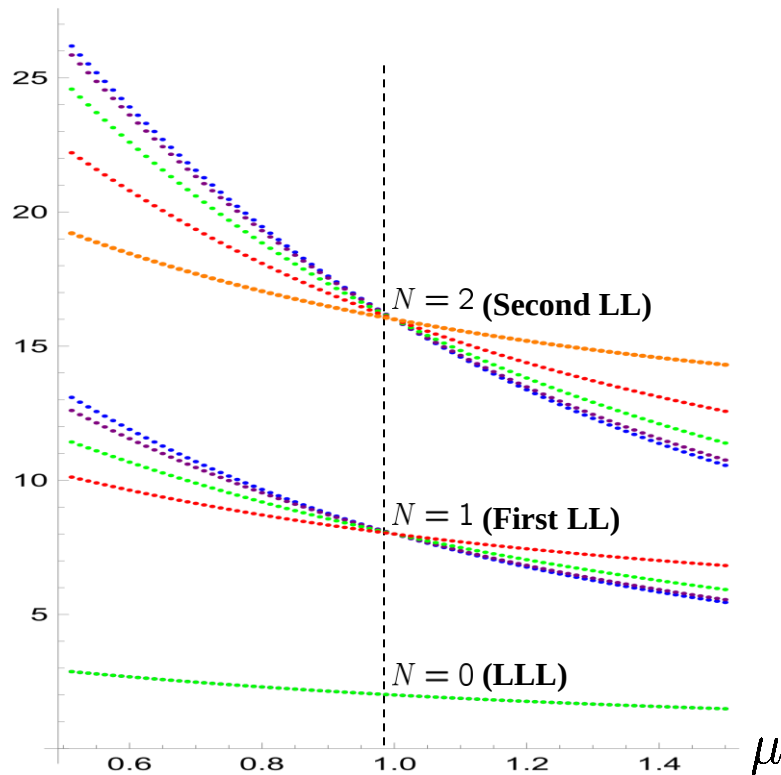
Deformed fuzzy geometry

7

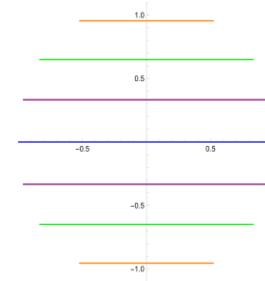
Spectral flow $\Rightarrow$ “Deformed” quantum states

$\Rightarrow$ Deformed fuzzy space (*KH* ‘25)

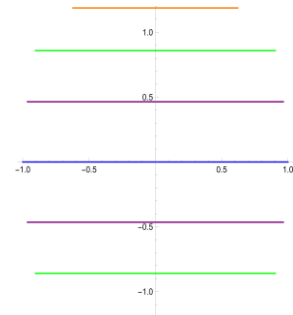
$$E^{(\mu)} \cdot 2M$$



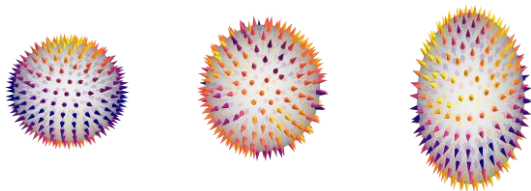
$\mu = 1.0$



$\mu = 1.4$



$\mu = 2.0$



(see also *Matsuura, Tsuchiya* ‘20)

Matrix geometry and Quantum states

8

Matrix model

(Matrix) Geometry
(vacuum)

True quantum vac.
(*Nishimura et al.*)

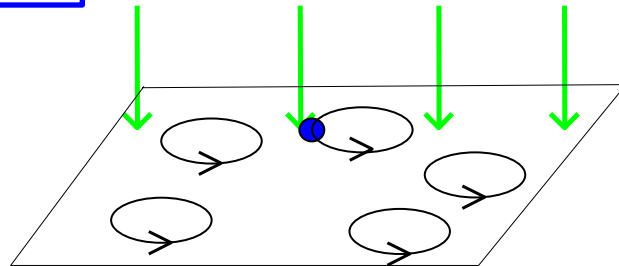


Fluctuations
(field theory)



Quantum states

Landau model



Level projection



Quantum states

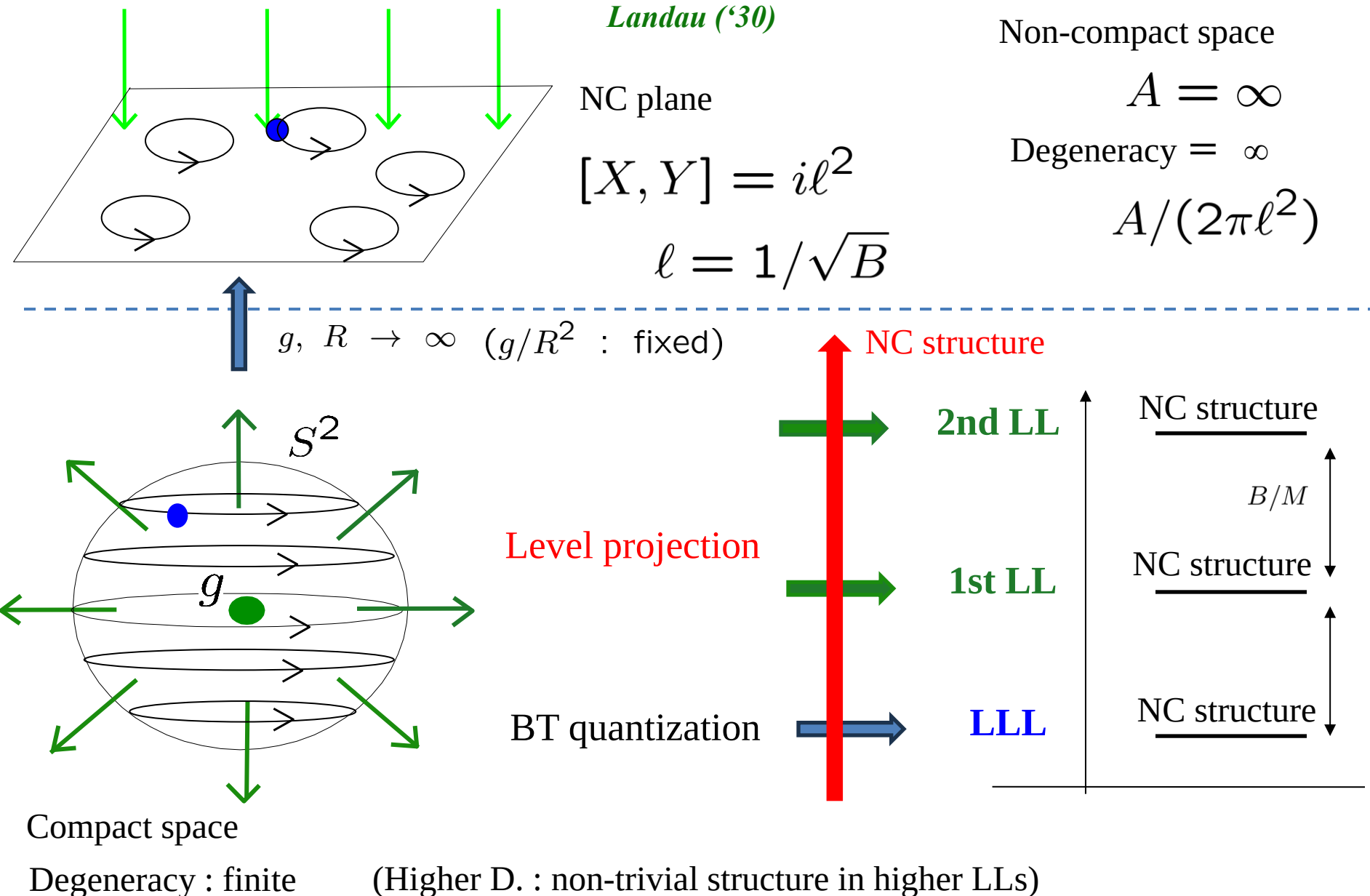


(Matrix) Geometry

*Hopefully,
not just conceptual, but
practically useful !*

The set-up

9



The Hopf maps and fuzzy spheres

10

The first Hopf map

$$S^3 \xrightarrow{S^1} S^2 \quad (\text{Hopf '31})$$

(Review *KH '10*)

$$\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

$$\rightarrow$$

$$x_i = \begin{pmatrix} \phi_1^* \\ \phi_2^* \end{pmatrix}^t \sigma_i \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

$$\phi_1^* \phi_1 + \phi_2^* \phi_2 = 1$$

$$x_i x_i = (\phi_1^* \phi_1 + \phi_2^* \phi_2)^2 = 1$$

The fuzzy sphere

From quantum state to geometry

$$[\hat{\phi}_1, \hat{\phi}_1^\dagger] = [\hat{\phi}_2, \hat{\phi}_2^\dagger] = 1$$

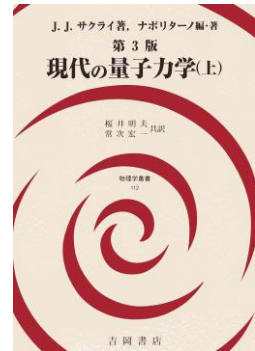
$$\rightarrow$$

$$X_i = \begin{pmatrix} \hat{\phi}_1^\dagger \\ \hat{\phi}_2^\dagger \end{pmatrix}^t \sigma_i \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix}$$

$$[\hat{\phi}_1, \hat{\phi}_2] = [\hat{\phi}_1, \hat{\phi}_2^\dagger] = 0$$

(Two independent quantum oscillators)

$$[X_i, X_j] = i\epsilon_{ijk} X_k$$



The second Hopf map, third Hopf map...

(*Hopf '31*)

The graded Hopf maps

(*Landi, Marmo '87.. KH '11*)

The non-compact Hopf maps

(*KH '10*)



Higher D fuzzy spheres

(*Grosse Klimcik Presnajder '95,...Kimura KH '04...*)



Fuzzy superspheres

(*Grosse Klimcik Presnajder '95,..., KH '11, '26*)



Higher D fuzzy hyperboloids

(*KH '12 ..*)

Non-Hermitian topological phase (*Esaki, Sato, KH, Kohmoto '11*), squeezed state (*KH '19*)..

Fuzzy sphere

Matrix coordinates *Hoppe ('82) Madore ('92)*

$$[X_i, X_j] = i\alpha\epsilon_{ijk}X_k$$

$$(X_i = \alpha S_i^{(l)})$$

$$\sum_{i=1}^3 X_i X_i = \alpha^2 l(l+1) 1_{2l+1}$$

Discrete but **continuous** rotational symmetry!

$$X_i \rightarrow R_{ij}(\theta, \phi) X_j$$

[cf.] *Snyder space-time ('47)*

「...理論が相対性理論と矛盾せず、...
彼の提出した理論に続くものがない」
(第12版1993年)

Matrix Model Solution

Myers ('99), Iso et al ('01), Kimura ('01, '02), Gohara Sako ('25)..

$$S = \text{tr}[X_i, X_j]^2 + X_i X_i$$

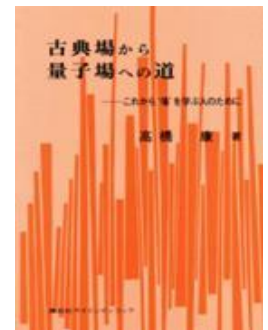
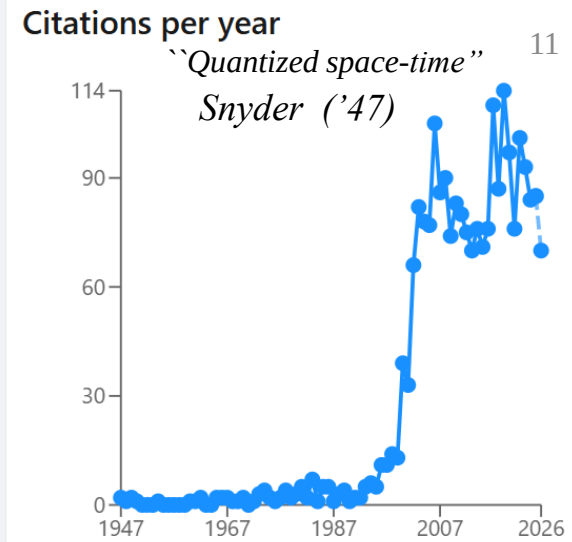
$$S = \text{tr}[X_i, X_j]^2 + \epsilon_{ijk} X_i X_j X_k$$

Numerical simulations of 3D Ising model (Fuzzy sphere regularization)

Zhu et al. ['22~] ...

<https://scgp.stonybrook.edu/archives/41863>

<https://indico.math.cnrs.fr/event/13856/>



Annual Review of Condensed Matter Physics

A Fuzzy Sphere Journey in
Critical Phenomena

Yin-Chen He¹ and W. Zhu²

¹ICNP, Xue Institute for Theoretical Physics, Stony Brook University, Stony Brook, New York

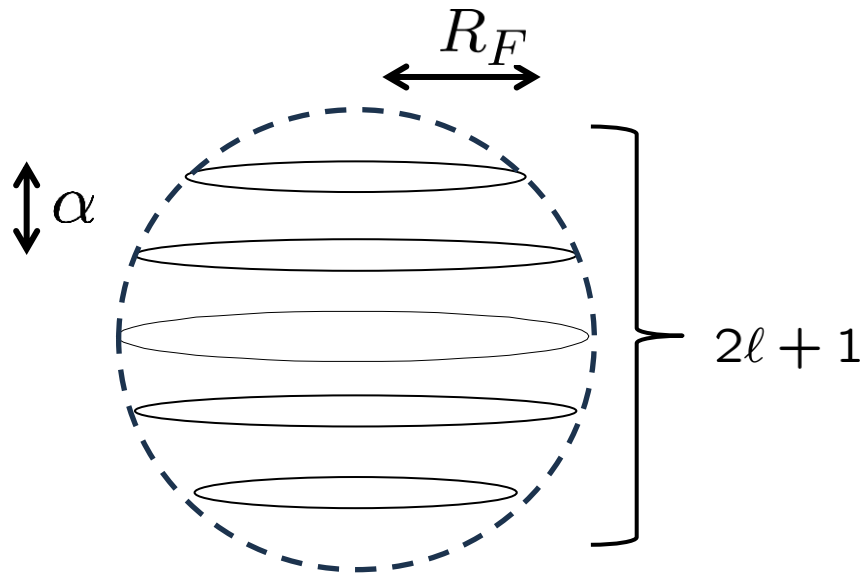
Intuitive picture of the fuzzy sphere

12

$$X_3 = \alpha \begin{pmatrix} \ell & 0 & 0 & 0 & 0 \\ 0 & \ell-1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & -(\ell-1) & 0 \\ 0 & 0 & 0 & 0 & -\ell \end{pmatrix}$$

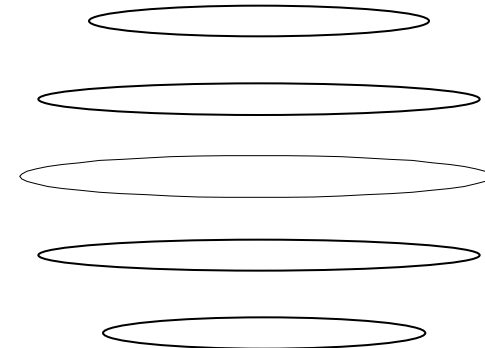
$$R_F = \sqrt{X_1^2 + X_2^2} = \sqrt{\mathbf{X}^2 - X_3^2}$$

$$= \alpha \begin{pmatrix} \sqrt{\ell} & 0 & 0 & 0 & 0 \\ 0 & \sqrt{3\ell+1} & 0 & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & \sqrt{3\ell+1} & 0 \\ 0 & 0 & 0 & 0 & \sqrt{\ell} \end{pmatrix}$$



Fuzzy sphere

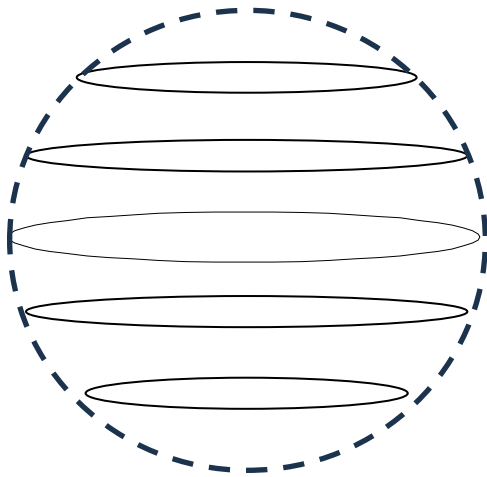
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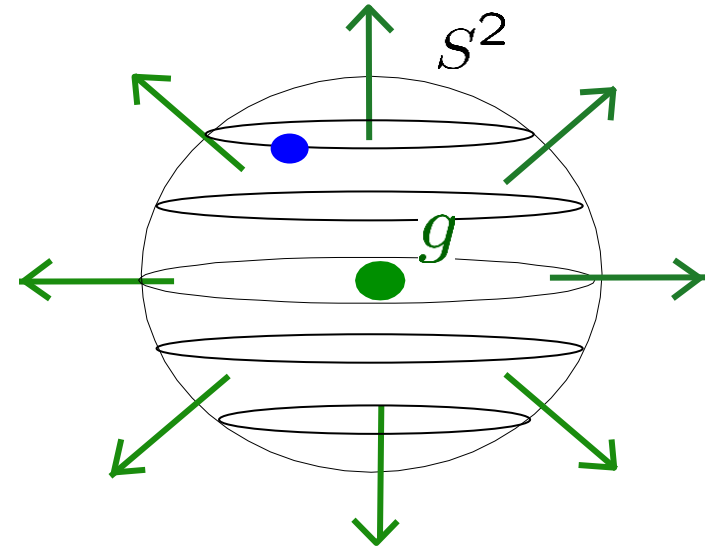
Quantum states
(irrep)

Quantum states => Fuzzy geometry

Physical model: Haldane's sphere



Fuzzy sphere



Haldane's sphere

Haldane ('83)

Monopole *Dirac ('31)*

$$A_i^{(g)} = -g \frac{1}{r(r + x_3)} \epsilon_{ij3} x_j$$

$$F_{ij}^{(g)} = g \frac{1}{r^3} \epsilon_{ijk} x_k$$

$$(g = 0, 1/2, 1, \dots)$$

Landau model on a sphere

14

Landau Hamiltonian on a two-sphere

Wu, Yang ('76) Haldane ('83) KH ('16)

$$H^{(g)} = -\frac{1}{2M} \sum_{i=1}^3 D_i^2 \Big|_{r=1} = \frac{1}{2M} \sum_{i=1}^3 \Lambda_i^2 \quad (\Lambda_i = -i\epsilon_{ijk}x_j(\partial_k + iA_k))$$

SU(2) angular momentum

$$L_i = \Lambda_i + g \frac{1}{r} x_i$$

SU(2) index

$$\ell = N + |g|$$

$$(N = 0, 1, 2, \dots)$$

Landau levels

$$E_N = \frac{1}{2M} (2|g|(N + \frac{1}{2}) + N(N + 1))$$

$$d_N = 2\ell + 1 = 2|g| + 2N + 1$$

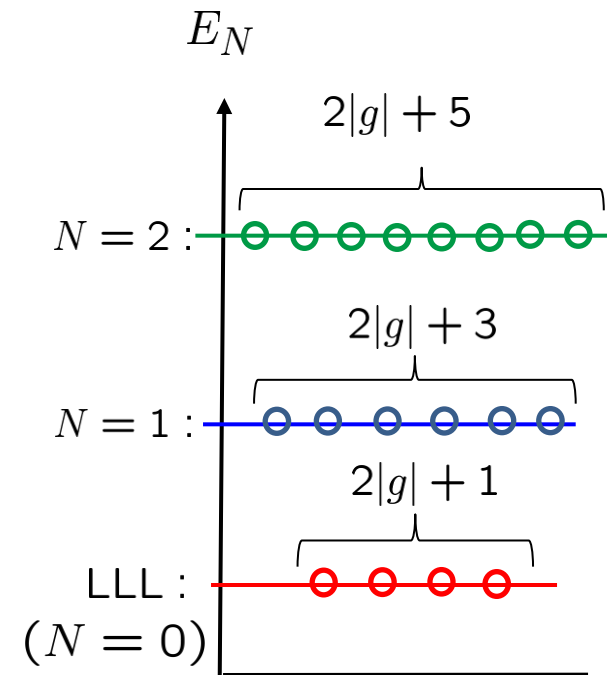
Eigenstates : monopole harmonics

$$Y_{l,m}^g(\theta, \phi)$$

$$(m = \ell, \ell - 1, \dots, -\ell)$$

NCG

$$[L_i, L_j] = i\epsilon_{ijk}L_k \quad \xrightarrow{\Lambda_i \simeq 0} \quad [X_i, X_j] \simeq i \frac{r}{g} \epsilon_{ijk} X_k$$



Emergence of fuzzy spheres from LLs

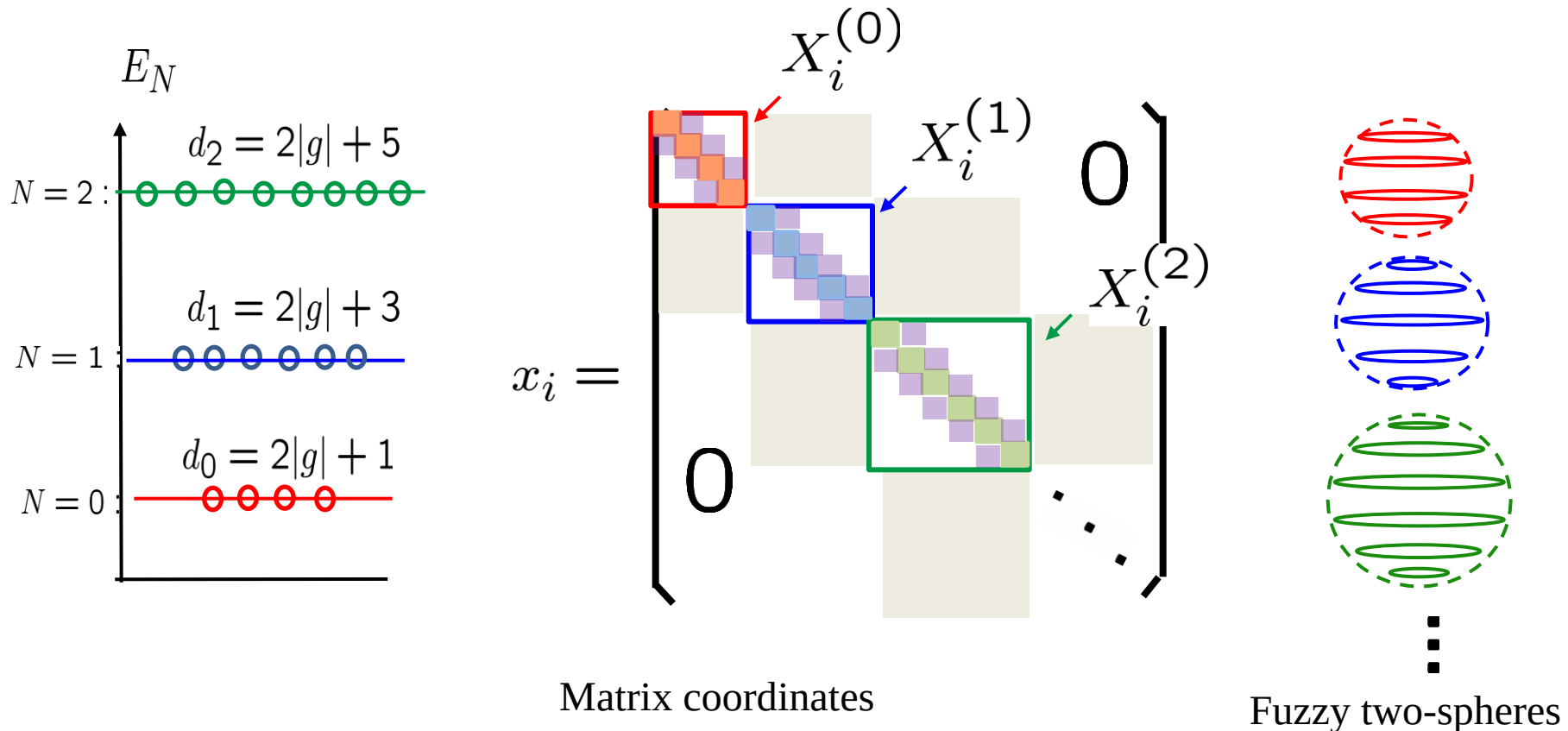
15

How can we precisely evaluate the LL geometries including higher ones?

KH [’16]

$$(X_i)_{m_1, m_2} = \langle Y_{l, m_1}^g | x_i | Y_{l, m_2}^g \rangle |_{S^2} = \frac{g}{l(l+1)} (S_i^{(l)})_{m_1, m_2}$$

$$\ell = N + |g|$$



Behind the scene

16

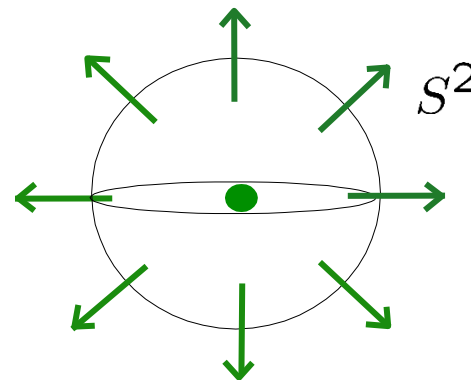
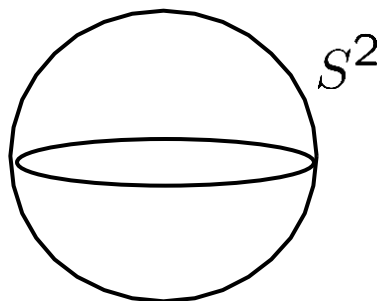
Reinterpretation

 S^2  S_F^2

- Global sym. of manifold, $SO(3)$
- Points on the manifold (= infinite dof)
: a closed set under the $SO(3)$ group action
- Stabilizer group $SO(2)$
= sym. that does not change a point

- Global sym. group of the system, $SU(2)$
- Irreps. (= finite dof) of $SU(2)$
: a closed set under the $SU(2)$ group action
- Gauge group $U(1)$
= sym. that does not change a physical state

$S^2 \simeq SO(3)/SO(2) \rightarrow$ QM with $SU(2)$ global sym. and $U(1)$ gauge sym.



(Magnetic field is *not* of primary importance; consequence of gauge sym.)

General prescription

17

$$\mathcal{M} \simeq G/H \longrightarrow \mathcal{M}_F$$

Just replace $SU(2) \rightarrow G$, and $U(1) \rightarrow H$.

KH ['23]

$\mathcal{M} \simeq G/H \longrightarrow$ QM with gauge symmetry H on the manifold $\mathcal{M}$

$$H = -\frac{1}{2M} \sum_i D_i^2 \Big|_{\mathcal{M}}$$

$\longrightarrow$ Solve the eigenvalue problem $\longrightarrow$ Matrix coordinates $\mathcal{M}_F$

$$|\psi_\alpha^{(N)}\rangle$$

$$(X_a^{(N)})_{\alpha\beta} = \langle \psi_\alpha^{(N)} | x_a | \psi_\beta^{(N)} \rangle$$

(Unique feature)

stabilizer group $\rightarrow$ gauge group

External space residual symmetry $\rightarrow$ **Internal space** gauge symmetry

Important Points

Quantum geometry is not postulated a priori,
but naturally emerges in context of physics.

- The original system is totally physical,
and its mathematical background is a consistent Hilbert space.
 - Hopefully, no need to worry about mathematical inconsistency

- Following the prescription, we can automatically construct the quantum matrix geometry corresponding to G/H .
 - Not restricted to symplectic manifolds; odd D is OK

- Not restricted to commutator formalism
 - Chance for exotic geometry such as Nambu geometry to appear

The monopole harmonics

19

$$Y_{l,m}^g(\theta, \phi)$$

$$\ell = N + |g|$$

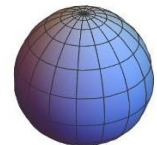
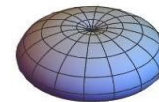
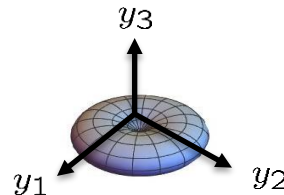
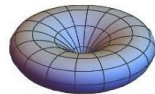
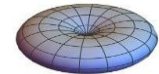
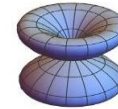
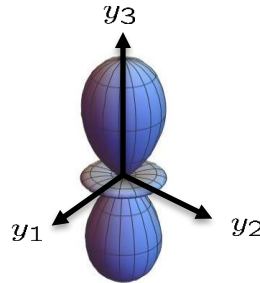
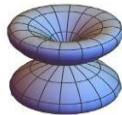
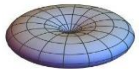
$$m = l, l-1, l-2, \dots, -l$$

$$g = l, l-1, l-2, \dots, -l$$

$$l = 2$$

Spherical harmonics (d-wave)

$$g = 0 \quad (N = 2)$$



$$g = 2 \quad (N = 0)$$

$$m : \quad 2$$

$$1$$

$$0$$

$$-1$$

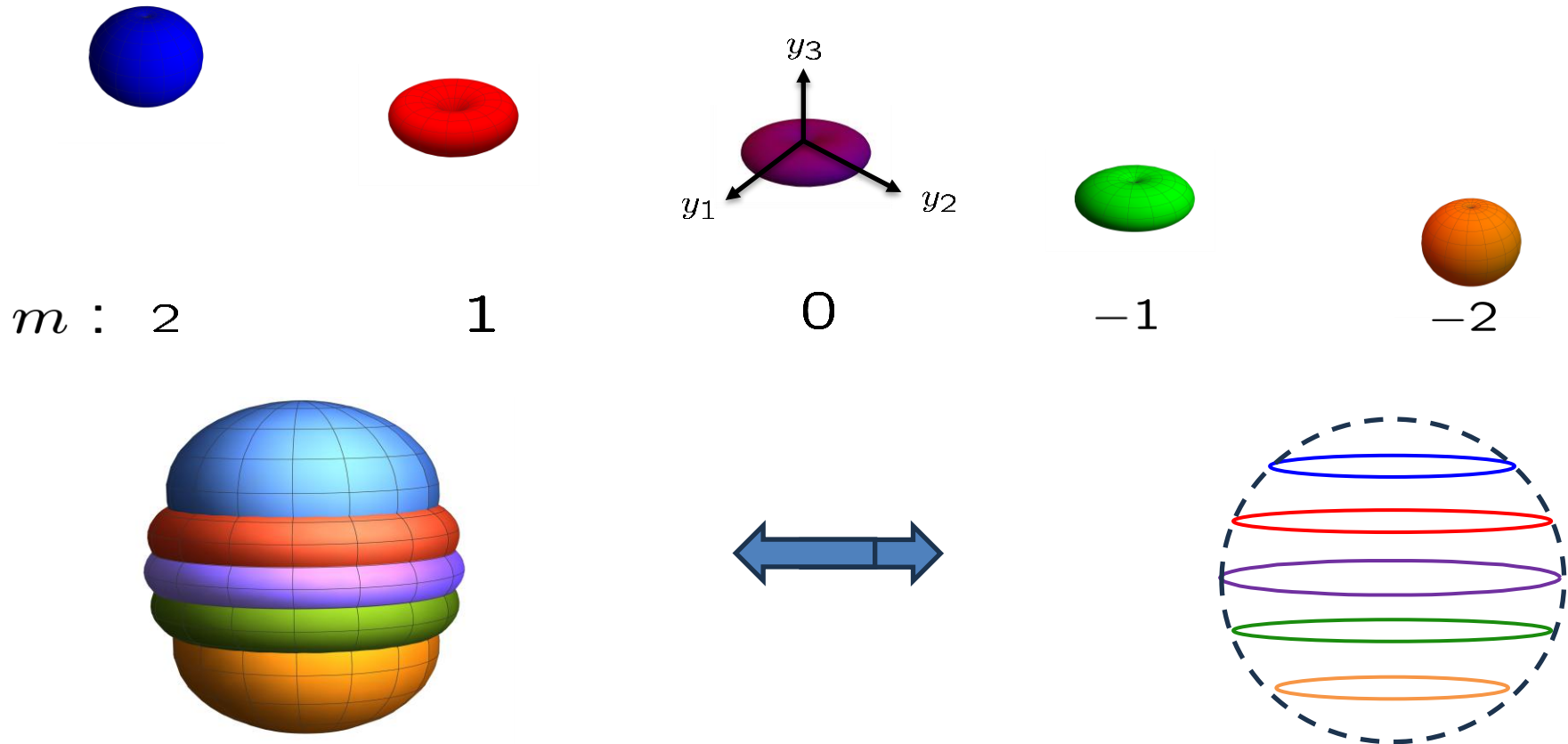
$$-2$$

LLL eigenstates

From quantum states to fuzzy sphere

20

LLL eigenstates ($l = 2$)



Simple understanding of fuzzy sphere geo. from quantum states !

Landau level eigenstates = $SU(2)$ irreps. = ``Points'' of fuzzy S^2

⇒ $SU(2)$ -symmetric NC space = $SU(2)$ irreps.

Irreps. as Quantum Geometry

21

$$\langle Y_{l,m_1}^g | x_i | Y_{l,m_2}^g \rangle \propto (S_i^{(l)})_{m_1, m_2} \quad (\text{obvious, from Wigner-Eckart theorem})$$

Irreps. is important

Let's forget about the Landau model.

Concentrate on the quantum states.

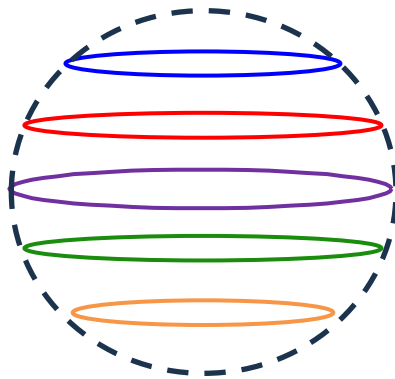
$$\{Y_{l,m}^g(\theta, \phi)\}$$

SU(2) rotation



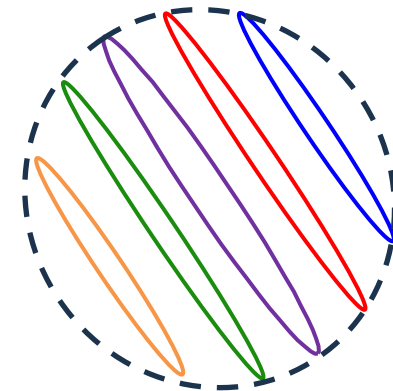
$$\{Y_{l,m}^g(\theta, \phi)\}$$

A set of a finite # of quantum states



SU(2) "invariant"

SU(2) rotation



A set of a finite # of quantum elements

SU(2) symmetry

Discrete (Finite number) but **continuous** rotational symmetry !

Quantum states (Irreps.) → Quantum Geometry

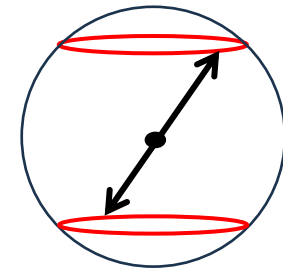
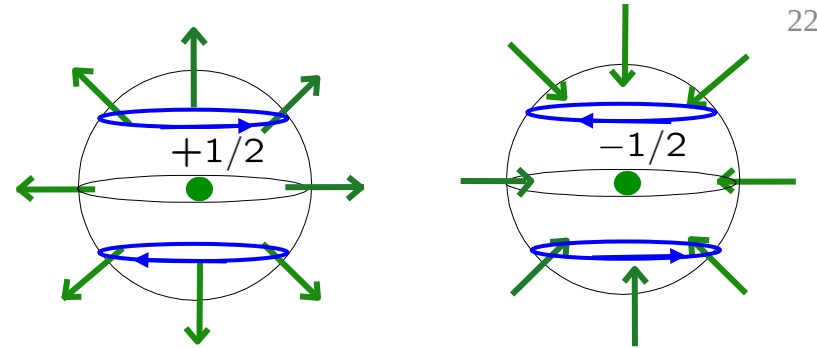
Monopole harmonics and spin-coherent states

$l = 1/2 \rightarrow 2\text{-fold degeneracy}$

$$(l = N + |g|) \quad g = \pm 1/2, N = 0$$

$$\begin{array}{cc}
 m = +1/2 & m = -1/2 \\
 g = +1/2 & \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} e^{i\phi} \\ -\sin \frac{\theta}{2} e^{i\phi} & \cos \frac{\theta}{2} \end{pmatrix} \\
 g = -1/2 & \\
 \end{array}
 = e^{i\frac{\chi}{2}\sigma_z} e^{i\frac{\theta}{2}\sigma_y} e^{i\frac{\phi}{2}\sigma_z} \Big|_{\chi=-\phi}$$

$S=1/2$ $S=-1/2$
 spin-coherent state spin-coherent state



$$\frac{1}{2} \boldsymbol{\sigma} \cdot \hat{\mathbf{x}}(\theta, \phi) \psi_S = S \psi_S$$

Row $\rightarrow$ degenerate LL eigenstates ($\rightarrow$ from classical geo to matrix geo.)

Column $\rightarrow$ (spin-)coherent state ($\rightarrow$ from matrix geo. to classical geo.)

$\rightarrow$ Ishiki ('15)

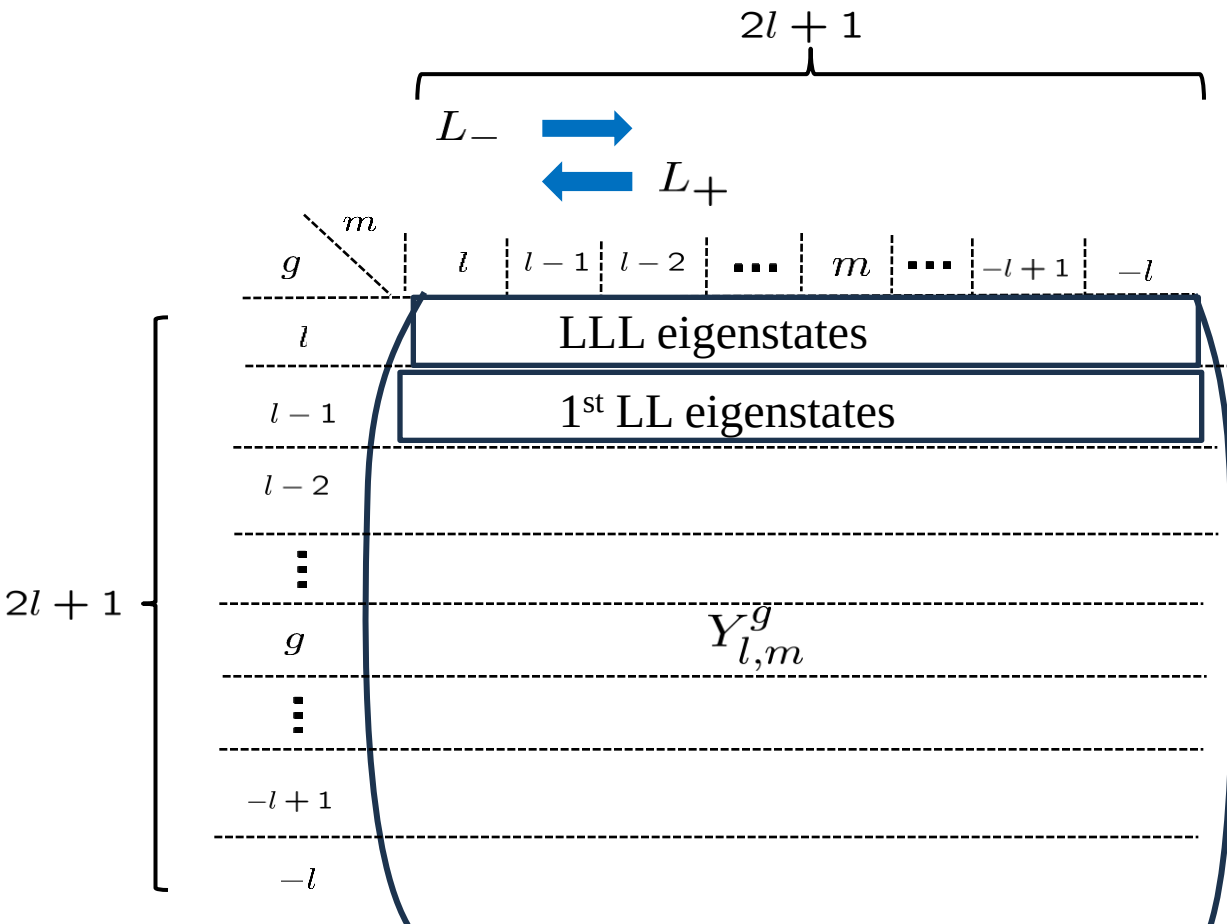
D-matrix as a unified description

23

$$Y_{l,m}^g(\theta, \phi) \quad m = l, l-1, l-2, \dots, -l \quad g = l, l-1, l-2, \dots, -l$$

Wu Yang('77) Dray('86) KH('16, '23)

l : fixed ($l = N + |g|$)



(D matrix)

$$= e^{i\chi S_z^{(l)}} e^{i\theta S_y^{(l)}} e^{i\phi S_z^{(l)}} \Big|_{\chi=-\phi}$$

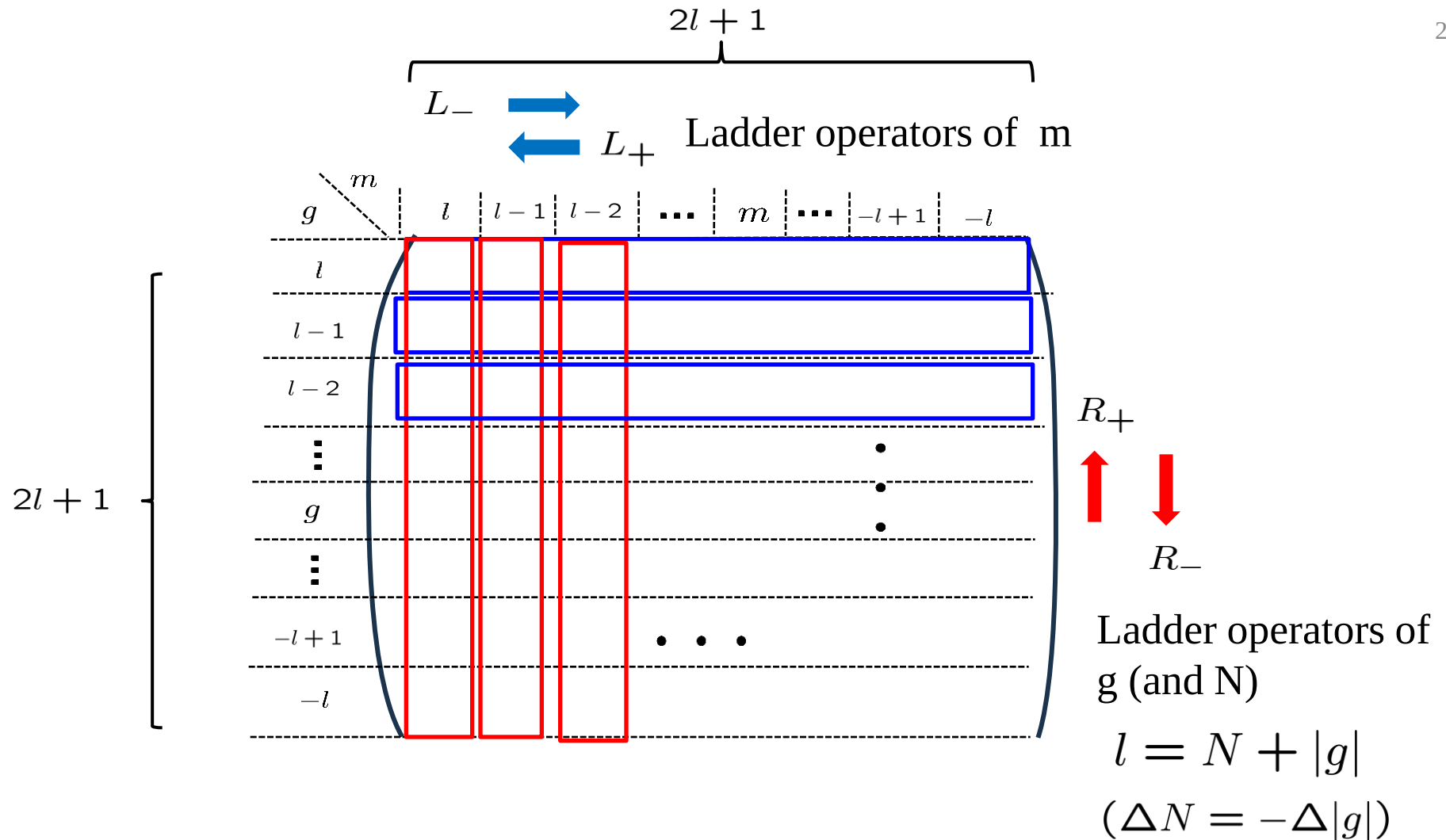
(= Euler angle rep. of SU(2))



* General feature of the Landau models on G/H

Two kinds of Ladder operators for D-matrix

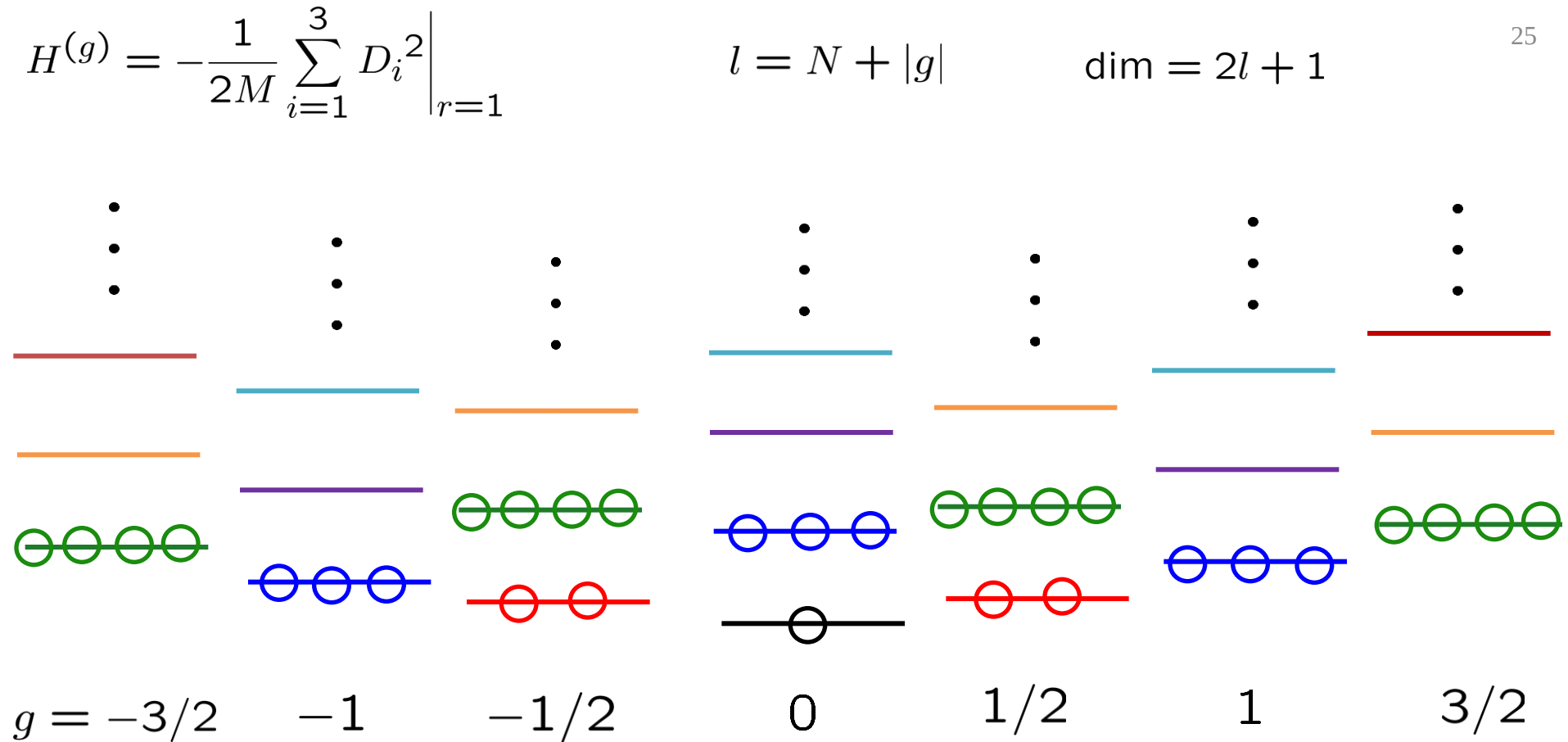
24



$$\mathcal{H} = \oplus_{l=0}^{\infty} \mathcal{H}^{(l)} \quad \text{Giant Hilbert space of a set of Hamiltonians } \{H^{(g)}\}$$

Howe duality from extended Landau levels

25



The degeneracy is exactly equal to the number of the corresponding LLs.

Howe duality is a hidden structure of a whole set of the Landau models.

(We cannot discover the Howe dual structure only by analyzing a single system.)

Howe correspondence

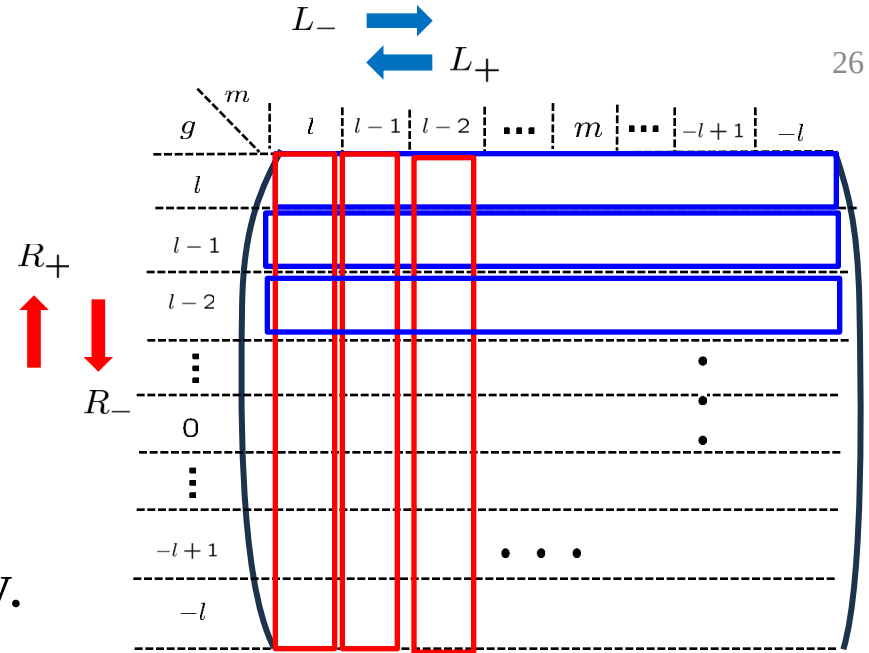
➤ Dual pair $(SU(2)_L, SU(2)_R)$

$$[L_i, L_j] = i\epsilon_{ijk}L_k \quad [L_i, R_j] = 0$$

$$[R_i, R_j] = i\epsilon_{ijk}R_k$$

Karabali Nair ['02] Greiter ['11]

$SU(2)_R$: spectrum generating algebra,
not a symmetry.



Momentum operators

(*Newman-Penrose*' edth operators [*'66*])

$$R_{\pm} \rightarrow \tilde{\partial}_{\pm}^{(g)} := D_{\theta} \pm i \frac{1}{\sin \theta} D_{\phi}$$

KH ['16]



R_i exactly coincide with the
covariant derivatives ∇_i
given by

Hanada, Kawai, Kimura (2005)

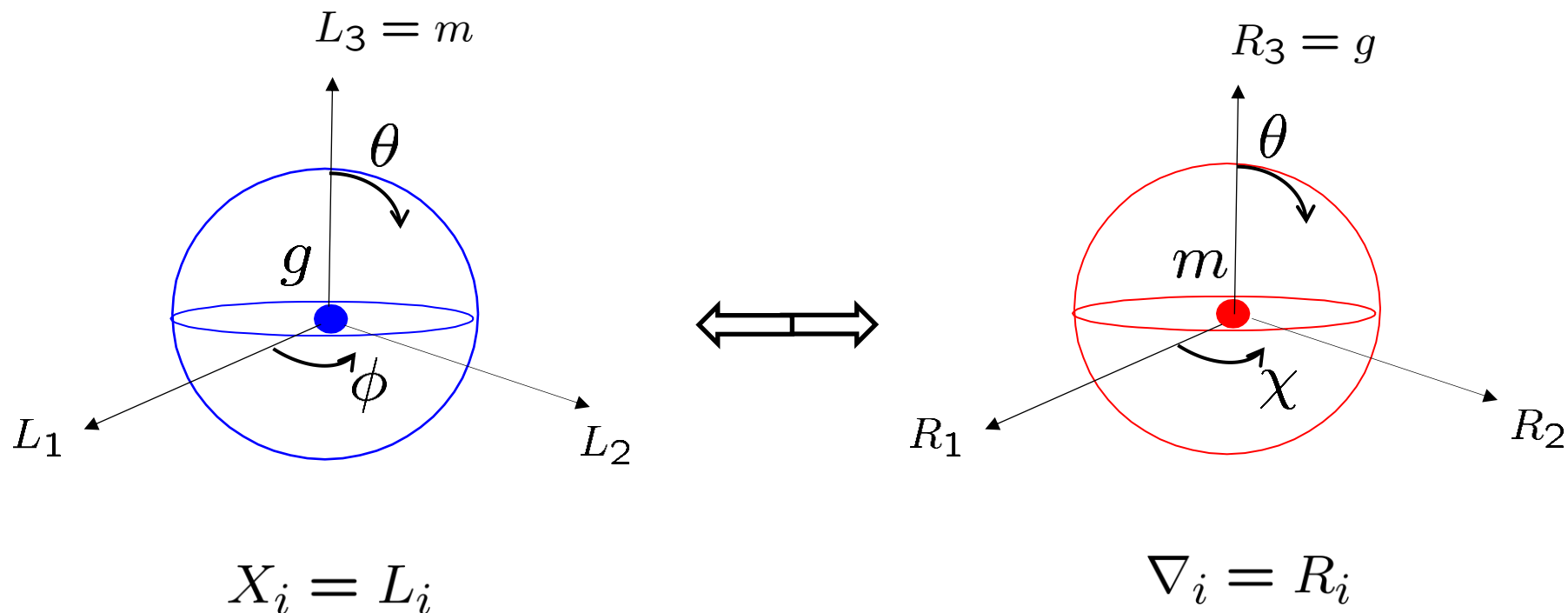
➤ Howe-correspondence

$$\mathcal{H} = \oplus_{l=0}^{\infty} (V_{SU(2)_L}^{(l)} \otimes V_{SU(2)_R}^{(l)}) \quad : \text{Howe duality}$$

Dual Landau model

27

$$L_i L_i = R_i R_i = l(l + 1)$$



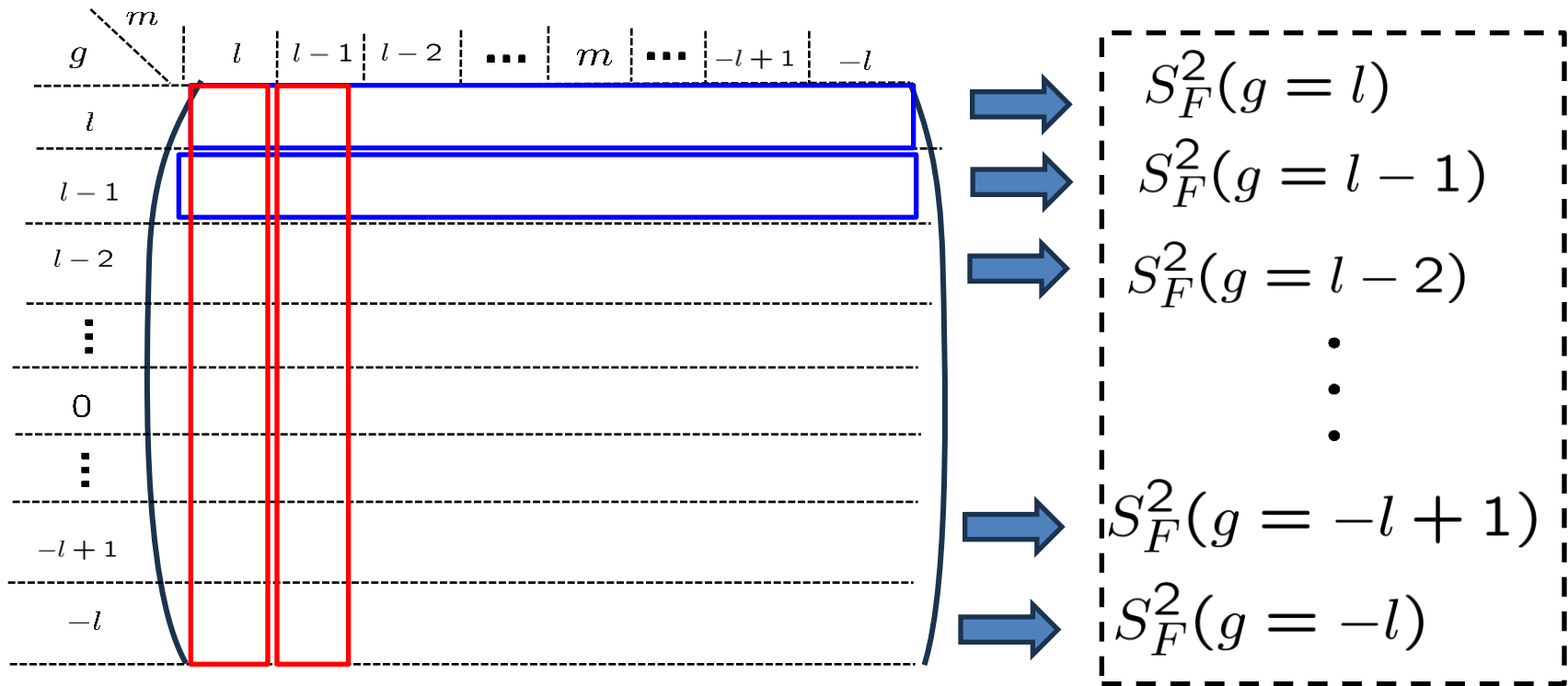
$$D_{l,g,m}(\chi, \theta, \phi) = \overbrace{(e^{i\chi S_z^{(l)}} e^{i\theta S_y^{(l)}} e^{i\phi S_z^{(l)}})}^{g,m} = e^{ig\chi} d_{l,g,m}(\theta) e^{im\phi}$$

Quantization of magnetic quantum number $\rightarrow$ topological charge in the dual system

Howe duality

$$l = N + |g|$$

A family of fuzzy spheres



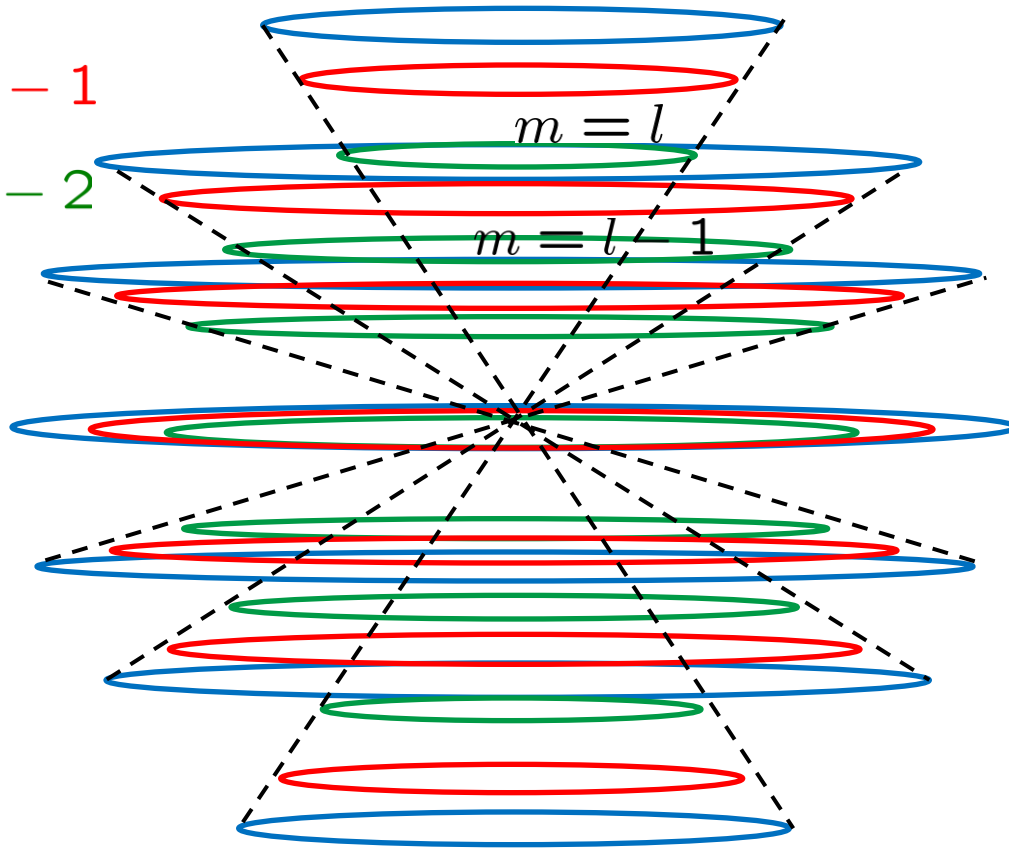
Howe duality
(Howe correspondence)

Emergence of dual fuzzy geometry

$$g = l$$

$$g = l - 1$$

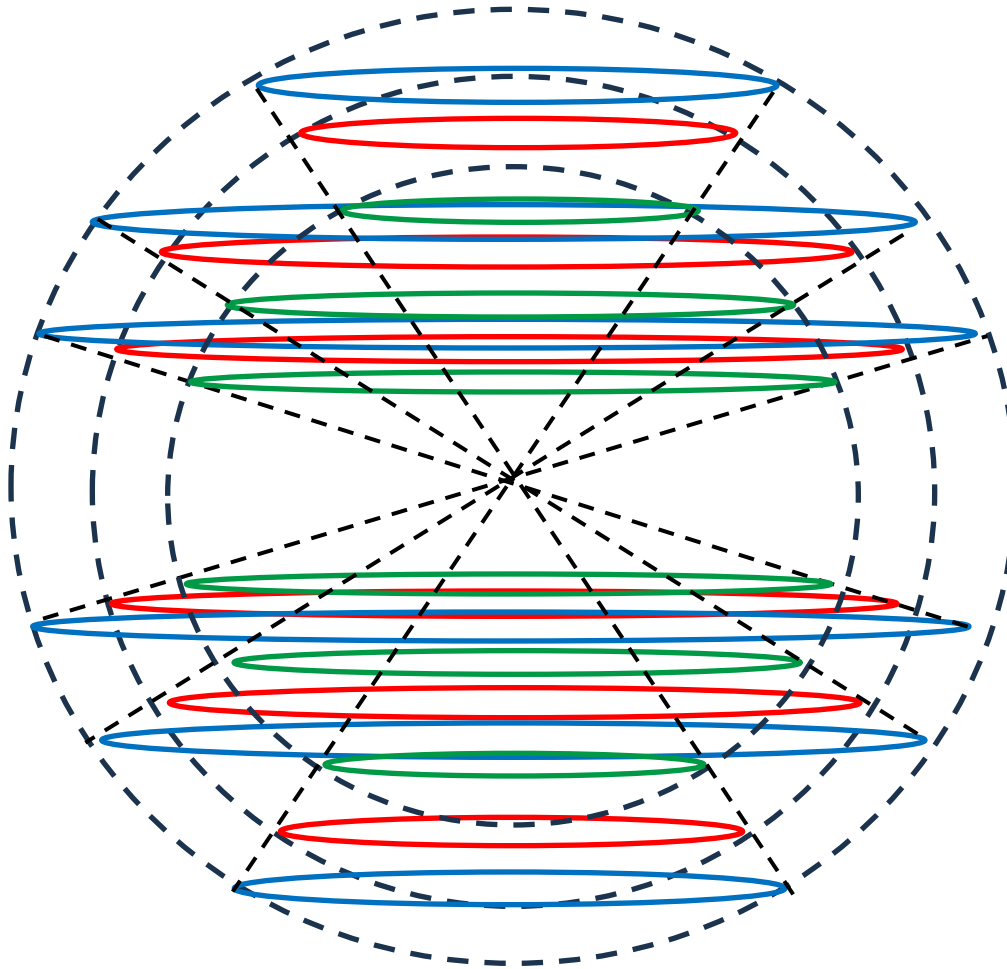
$$g = l - 2$$



Dual geometry emerges once the *whole structure* is identified.

Intuitive understanding of the duality

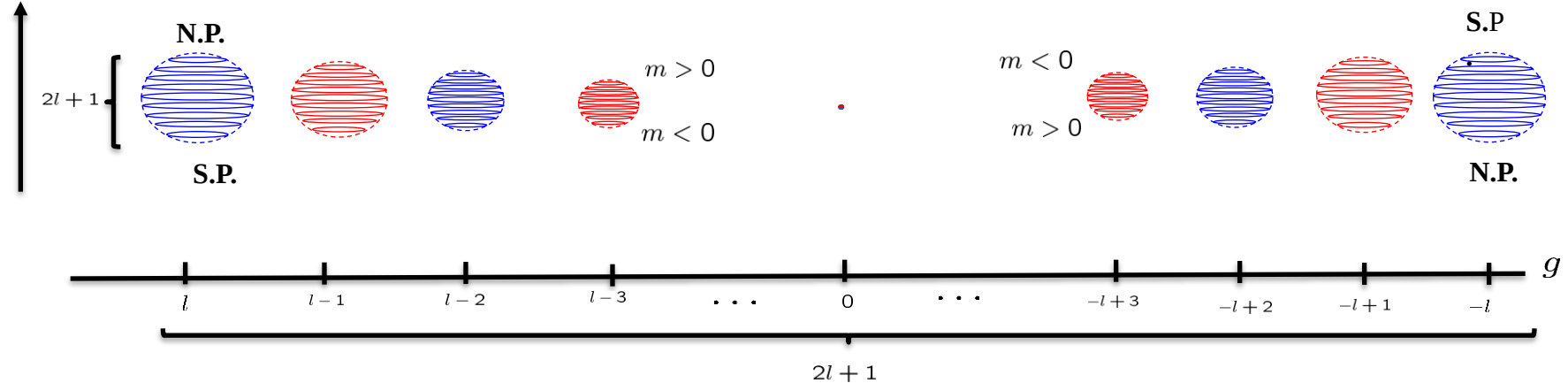
30



Two faces of single object

Howe correspondence in NCG

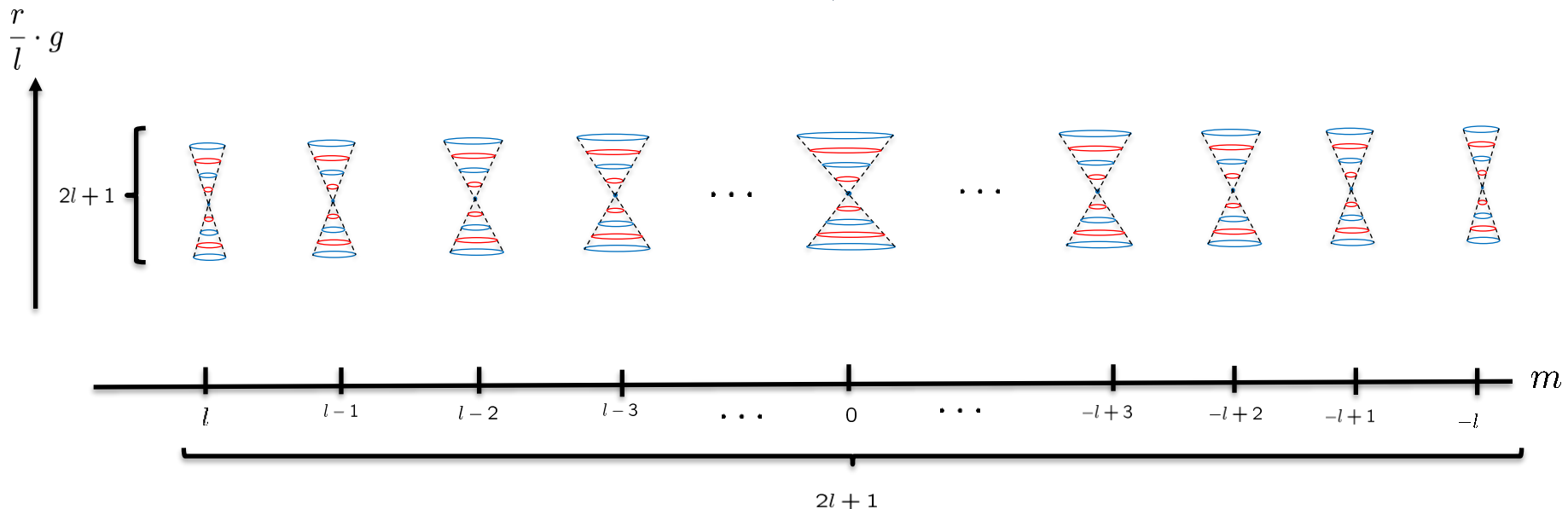
$$X_3 = \alpha \cdot m$$



$$X_i = \frac{g}{l(l+1)} S_i^{(l)}$$



Howe correspondence



Internal and external spaces in Matrix geo.

Fuzzy S4 coordinates

$SO(5) \quad \gamma_a=1,2,3,4,5$

$$[\gamma_a, \gamma_b] = i\sigma_{ab}$$

External space

Internal space

(Ho-Ramgoolam 2002, Kimura 2002, Kimura-KH 2004)

common algebraic origin, same matrix size

$$\gamma_a \longrightarrow X_a$$

$$\sigma_{ab} \longrightarrow \Sigma_{ab}$$

→ *Higher spin structure from IKKT model*

(Steinacker 2016,2019,2023)

Internal space $\simeq$ External space

momentum A_μ *space-time coordinates*

$(\nabla_i \Leftrightarrow X_i)$ *(Hanada, Kawai, Kimura 2005,*

Ho, Kawai, Steinacker 2026)

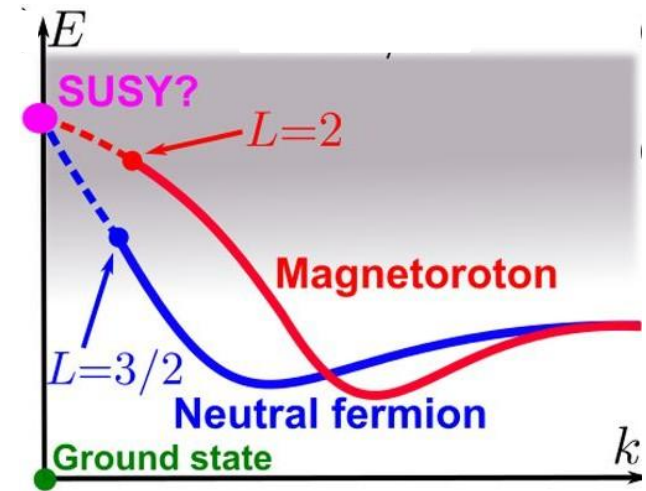
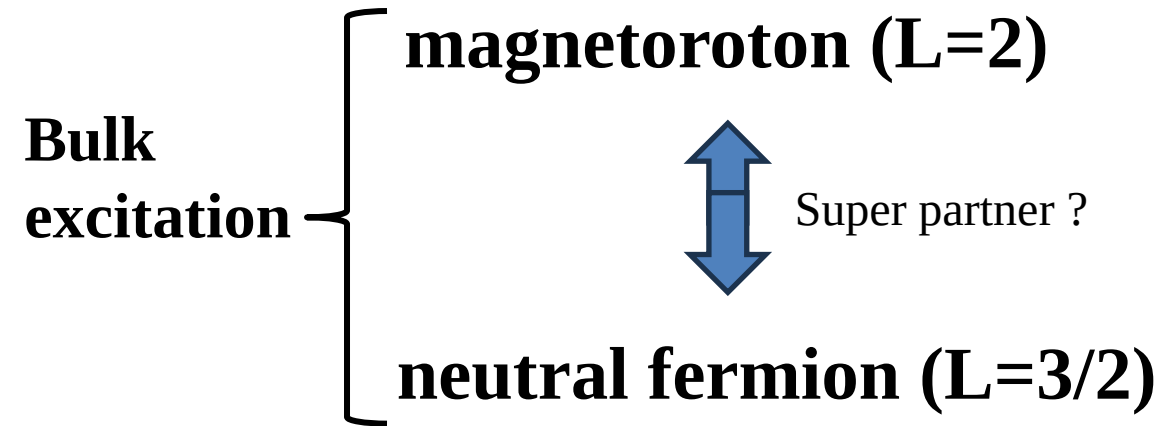
→ *Howe duality ?*

Renewed interest in SUSY NCG

33

Moore-Read state (even denominator QHE)

$$\nu = \frac{5}{2}$$



*Pu, Balram, et al ('23) PRL
130, 176501*

Gromov, Martinec, Ryu ('20), Pu, Balram, Femling, Gromov, Papic ('23), ...

[supersymmetry in fractional quantum Hall systems -- references in nLab](#)

Exact SUSY of the fuzzy supersphere

34

supersphere

$$x_i x_i + \epsilon_{\alpha\beta} \theta_\alpha \theta_\beta = R^2$$

$\mathcal{N} = 1$ SUSY (UOSP(1|2) group)

$$\left\{ \begin{array}{l} \delta_\eta x_i = -\xi_\alpha \frac{1}{2} (\sigma_i)_{\beta\alpha} \theta_\beta \\ \delta_\eta \theta_\alpha = i \xi_\beta \frac{1}{2} (\sigma_2 \sigma_i)_{\beta\alpha} x_i \end{array} \right.$$

Fuzzy supersphere

Grosse, Klimcik, Presnajder ('97)

Grosse, Reiter ('98)

Iso, Umetsu ('04)

$$X_i X_i + \epsilon_{\alpha\beta} \Theta_\alpha \Theta_\beta \propto \mathbf{1}$$

Exact $\mathcal{N} = 1$ SUSY

$$\left\{ \begin{array}{l} \delta_\eta X_i = -\xi_\alpha \frac{1}{2} (\sigma_i)_{\beta\alpha} \Theta_\beta \\ \delta_\eta \Theta_\alpha = i \xi_\beta \frac{1}{2} (\sigma_2 \sigma_i)_{\beta\alpha} X_i \end{array} \right.$$

$$[X_i, X_j] = i \epsilon_{ijk} X_k$$

$$[X_i, \Theta_\alpha] = \frac{1}{2} (\sigma_i)_{\beta\alpha} \Theta_\beta$$

$$\{\Theta_\alpha, \Theta_\beta\} = i \frac{1}{2} (\sigma_2 \sigma_i)_{\alpha\beta} X_i.$$

uosp(1|2) algebra

Super Landau model

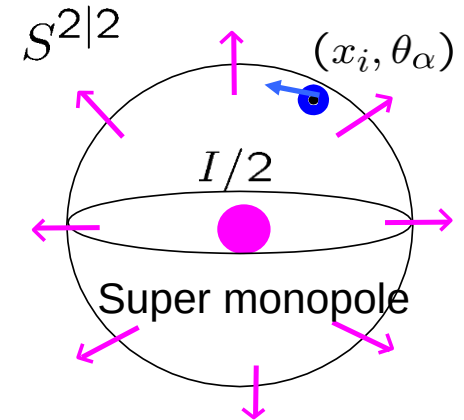
35

Super Landau model

Kimura, KH ('04) KH ('26)

$$H = \frac{1}{2MR^2}(\Lambda_i \Lambda_i + \epsilon_{\alpha\beta} \Lambda_\alpha \Lambda_\beta)$$

on supersphere $x_i x_i + \epsilon_{\alpha\beta} \theta_\alpha \theta_\beta = R^2$



$$\begin{cases} \Lambda_a = -i\epsilon_{abc}x_b D_c + \frac{1}{2}\theta_\alpha(\sigma_a)_{\alpha\beta}D_\beta & A_i = g\epsilon_{ij3}\frac{x_j}{r(r+x_3)}(1 + \frac{r+x_3}{2r^2(r+x_3)}\theta\epsilon\theta) \\ \Lambda_\alpha = \frac{1}{2}(\epsilon\sigma_a)_{\alpha\beta}x_a D_\beta - \frac{1}{2}\theta_\beta(\sigma_a)_{\beta\alpha}D_a & A_\alpha = g\frac{1}{r^3}i(\sigma_i\epsilon)_{\alpha\beta}x_i\theta_\beta \end{cases}$$

$$E_N = \frac{1}{2MR^2}(N(N + \frac{1}{2}) + (2N + \frac{1}{2}|g|))$$

$$N = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$$

$$d_N = 4l + 1$$

$$l = N + |g|$$

Supermonopole Harmonics

36

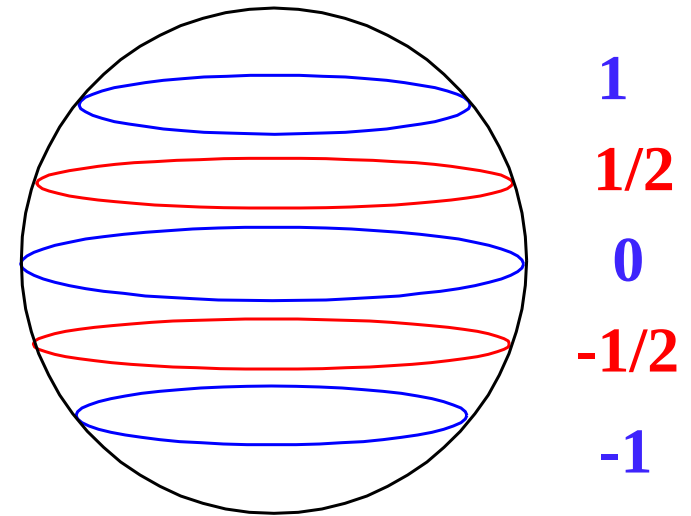
$UOSp(1|2)$ superspin $l = 0, 1/2, 1, 3/2, \dots$ (Ex.) $l = 1$

→ $SU(2) \quad l \oplus l - \frac{1}{2}$

$$d = \underbrace{(2l + 1)}_{\text{Bosonic d.o.f.}} + \underbrace{(2l)}_{\text{Fermionic d.o.f.}} = 4l + 1$$

Bosonic d.o.f. Fermionic d.o.f.

$SU(2) \quad l = 1 \quad l = 1/2 \quad m$



LLL

$$Y_{m_1, m_2} = \phi_1^{m_1} \phi_2^{m_2},$$

$$\mathcal{Y}_{n_1, n_2} = \phi_1^{n_1} \phi_2^{n_2} \eta \quad (\eta := \phi_1 \theta_2 - \phi_2 \theta_1)$$

$$m_1 + m_2 = 2l \quad n_1 + n_2 = 2l - 1$$

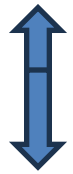
(No complex conjugate variables)

The SUSY system is (almost) doubly degenerate due to the existence of the fermionic d.o.f.

Super Landau Levels and Howe duality ³⁷

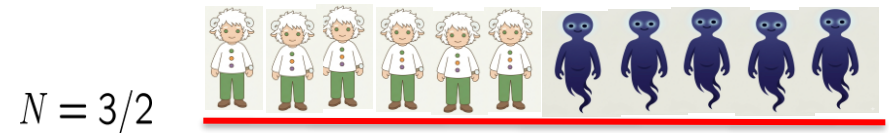
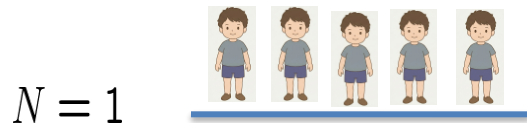
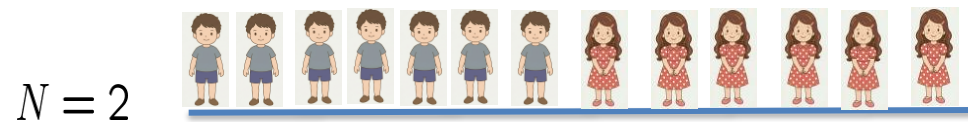
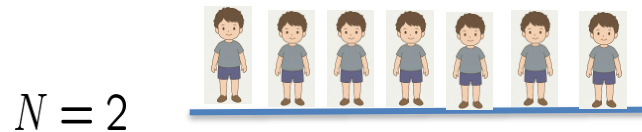
Supermonopole harmonics =

bosonic monopole harmonics + fermionic monopole harmonics

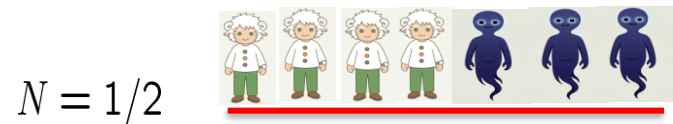


Super Howe duality

Super LLs = Integer LLs + Half-integer LLs



SUSY

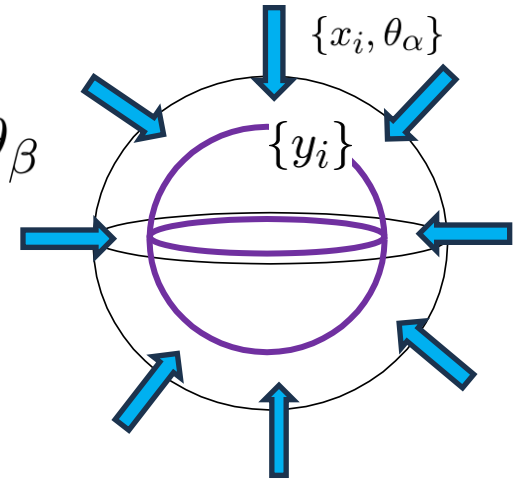


Quantum Mechanics on a supermanifold

38

$$x_i x_i + \epsilon_{\alpha\beta} \theta_\alpha \theta_\beta = R^2 \quad \Rightarrow \quad x_i x_i = R^2 - \epsilon_{\alpha\beta} \theta_\alpha \theta_\beta$$

$$y_i y_i = R^2 \quad \text{body-sphere} \quad x_i = \sqrt{1 - \frac{1}{R^2} \epsilon_{\alpha\beta} \theta_\alpha \theta_\beta} y_i$$



➤ Projection to the body

$$\rho(y) = \int d\theta_1 d\theta_2 \left(1 - \frac{1}{2} \epsilon_{\alpha\beta} \theta_\alpha \theta_\beta\right) \Psi(x, \theta)^* \Psi(x, \theta)$$

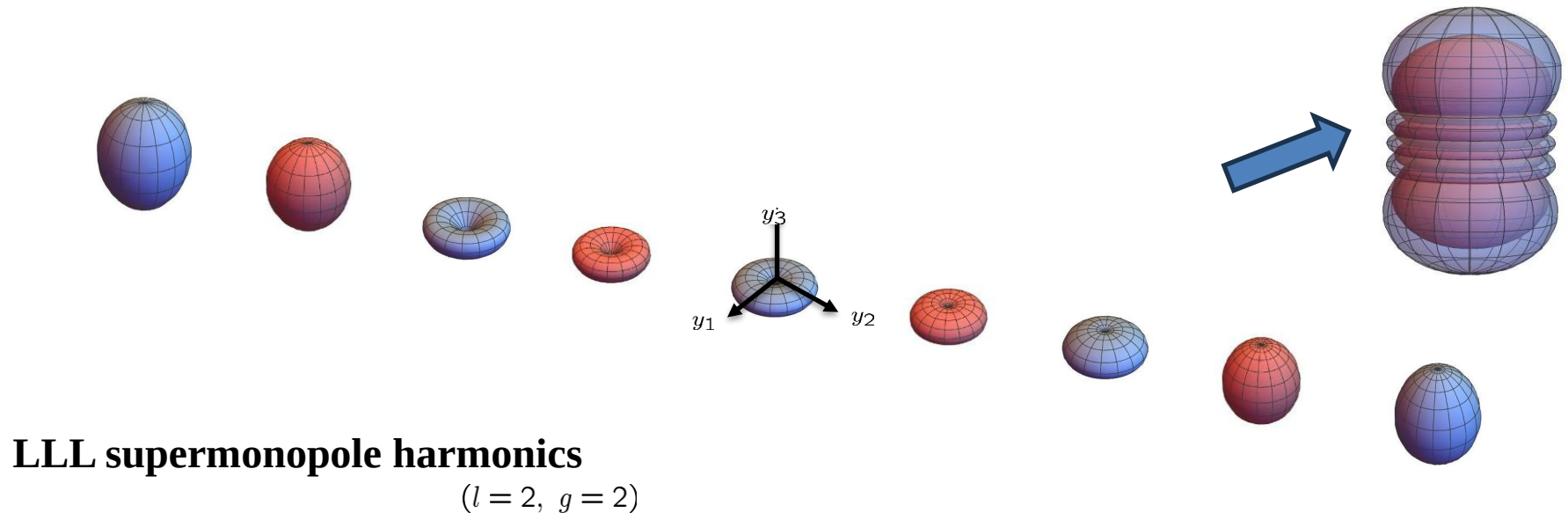
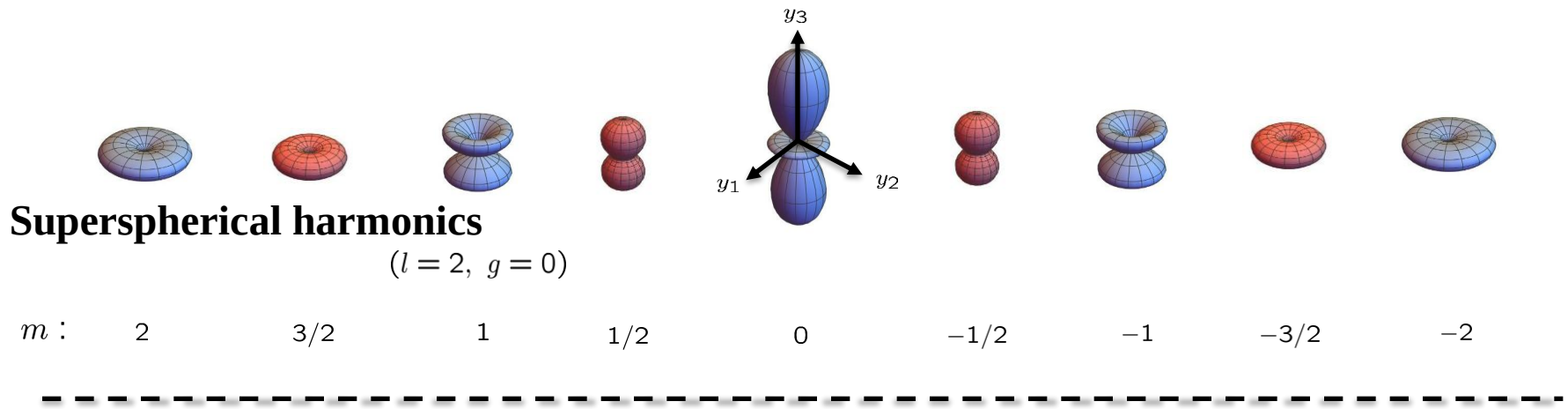
➤ Appropriate inner product (redefinition of the metric in SUSY Hilbert space)

$$\int d\theta_1 d\theta_2 \left(1 - \frac{1}{2} \theta \epsilon \theta\right) \int_{S^2} d\Omega_2 \Psi_{\text{Ghost}}^* \Psi_{\text{Ghost}} = -1$$

$$\langle \Psi_{\text{Ghost}} | \equiv -\Psi_{\text{Ghost}}^*$$

The supermonopole harmonics

39



Super quantum matrix geometry

40

“Bosonic” monopole harmonics

“Fermionic” monopole harmonics

$$Y_{l,m}^g(x_i, \theta_\alpha)$$

$$\mathcal{Y}_{l,m}^g(x_i, \theta_\alpha)$$

$$X_i := \begin{pmatrix} \langle Y_l | x_i | Y_l \rangle & \langle Y_l | x_i | \mathcal{Y}_l \rangle \\ \langle \mathcal{Y}_l | x_i | Y_l \rangle & \langle \mathcal{Y}_l | x_i | \mathcal{Y}_l \rangle \end{pmatrix} \quad \Theta_\alpha := \begin{pmatrix} \langle Y_l | x_i | Y_l \rangle & \langle Y_l | x_i | \mathcal{Y}_l \rangle \\ \langle \mathcal{Y}_l | x_i | Y_l \rangle & \langle \mathcal{Y}_l | x_i | \mathcal{Y}_l \rangle \end{pmatrix}$$

: $(4l + 1) \times (4l + 1)$ matrices

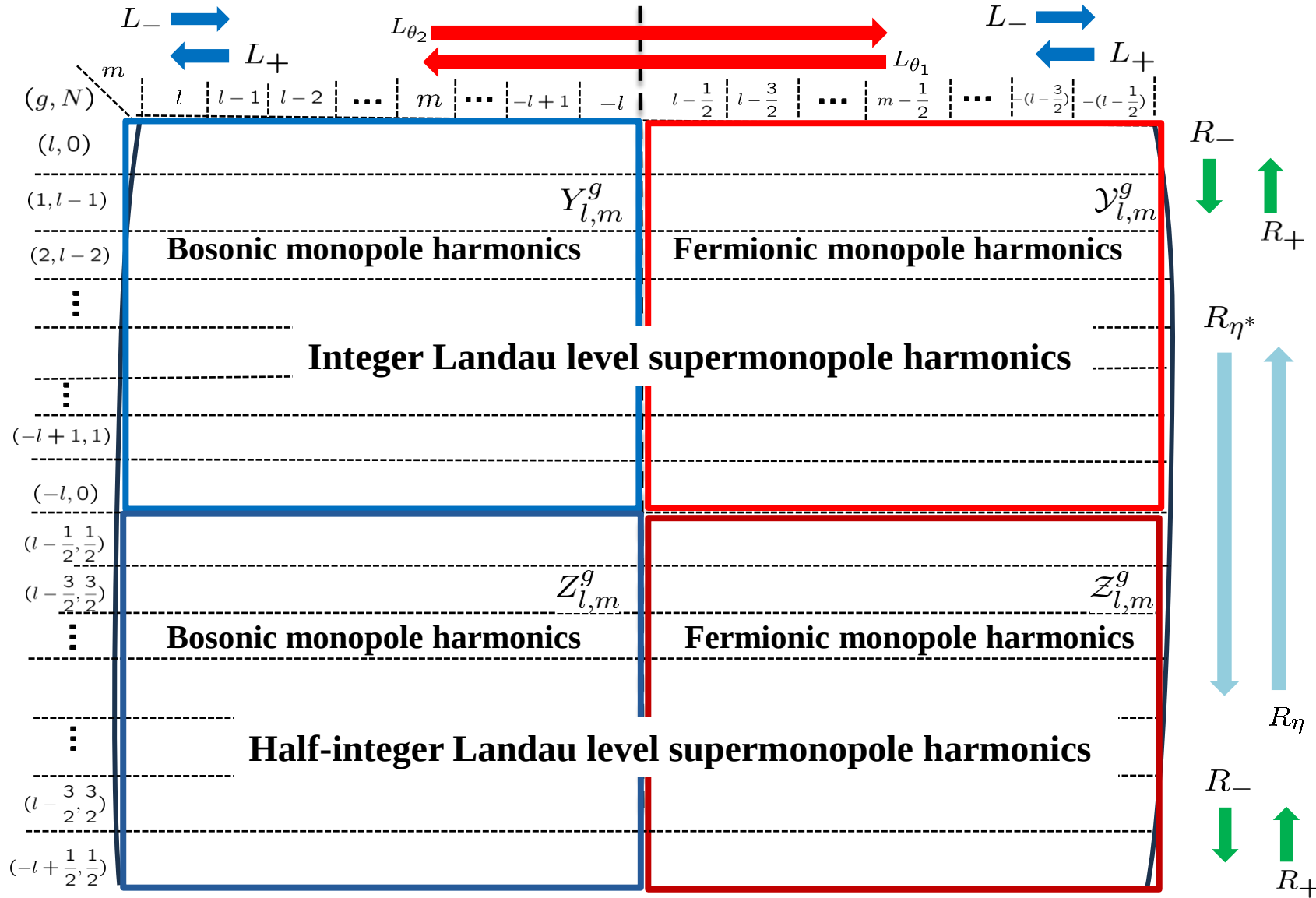
$$\text{Fuzzy supersphere algebra} \left\{ \begin{array}{l} [X_i, X_j] = i\alpha \epsilon_{ijk} X_k \\ [X_i, \Theta_\alpha] = \alpha \frac{1}{2} (\sigma_i)_{\beta\alpha} \Theta_\beta \\ \{\Theta_\alpha, \Theta_\beta\} = i\alpha \frac{1}{2} (\sigma_2 \sigma_i)_{\alpha\beta} X_i. \end{array} \right. \quad \alpha := \frac{g}{l(l + 1/2)}$$

$$X_i X_i + \epsilon_{\alpha\beta} \Theta_\alpha \Theta_\beta = \alpha^2 l(l + 1/2) 1_{4l+1}$$

Super D-matrix

41

$$\mathcal{D} = e^{-i\chi L_z} e^{-i\theta L_y} e^{-i\phi L_z} e^{2\eta L_{\theta_1} + 2\eta^* L_{\theta_2}}$$



Super Howe duality

42

➤ Dual pair $(UOSp(1|2)_L, UOSp(1|2)_R)$

$$\left\{ \begin{array}{l} [L_i, L_j] = i\epsilon_{ijk}L_k \\ [L_i, L_\alpha] = \frac{1}{2}(\sigma_i)_{\beta\alpha}L_\beta \\ \{L_\alpha, L_\beta\} = i\frac{1}{2}(\sigma_2\sigma_i)_{\alpha\beta}L_i. \end{array} \right. \quad \left\{ \begin{array}{l} [R_i, R_j] = i\epsilon_{ijk}R_k \\ [R_i, R_\alpha] = \frac{1}{2}(\sigma_i)_{\beta\alpha}R_\beta \\ \{R_\alpha, R_\beta\} = i\frac{1}{2}(\sigma_2\sigma_i)_{\alpha\beta}R_i. \end{array} \right.$$

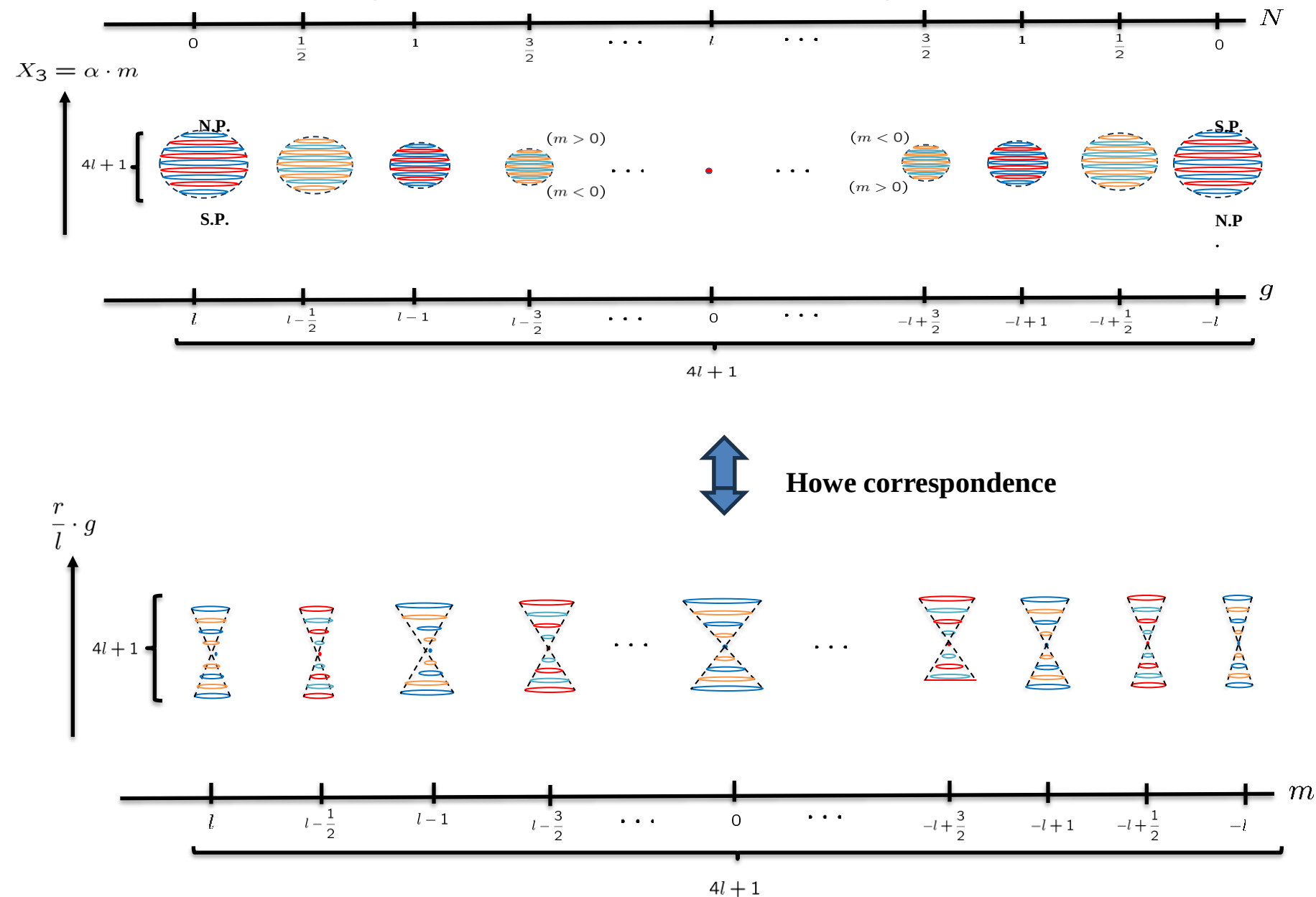
$$[L_i, R_j] = [L_i, R_\alpha] = [L_\alpha, R_i] = \{L_\alpha, R_\beta\} = 0$$

➤ Howe-correspondence

$$\mathcal{H} = \oplus_{l=0}^{\infty} (V_{UOSp(1|2)_L}^{(l)} \otimes V_{UOSp(1|2)_R}^{(l)})$$

The Super Howe correspondence

43



Weinberg の著書 より

44

S. Weinberg, ``Dream of a Final Theory'' 1994

「究極理論への夢」 小野十加藤 訳 ダイヤモンド社



第6章 美しい理論

物理理論の美しさは基礎の簡単な原理の上につくられた、
剛い数学的構造の中に具現化されるが、不思議なことに、
この種類の美しさをもつ構造は、基礎の原理が誤りと分かった後
も生きながらえる傾向にある。よい例はディラックの電子論である。



....

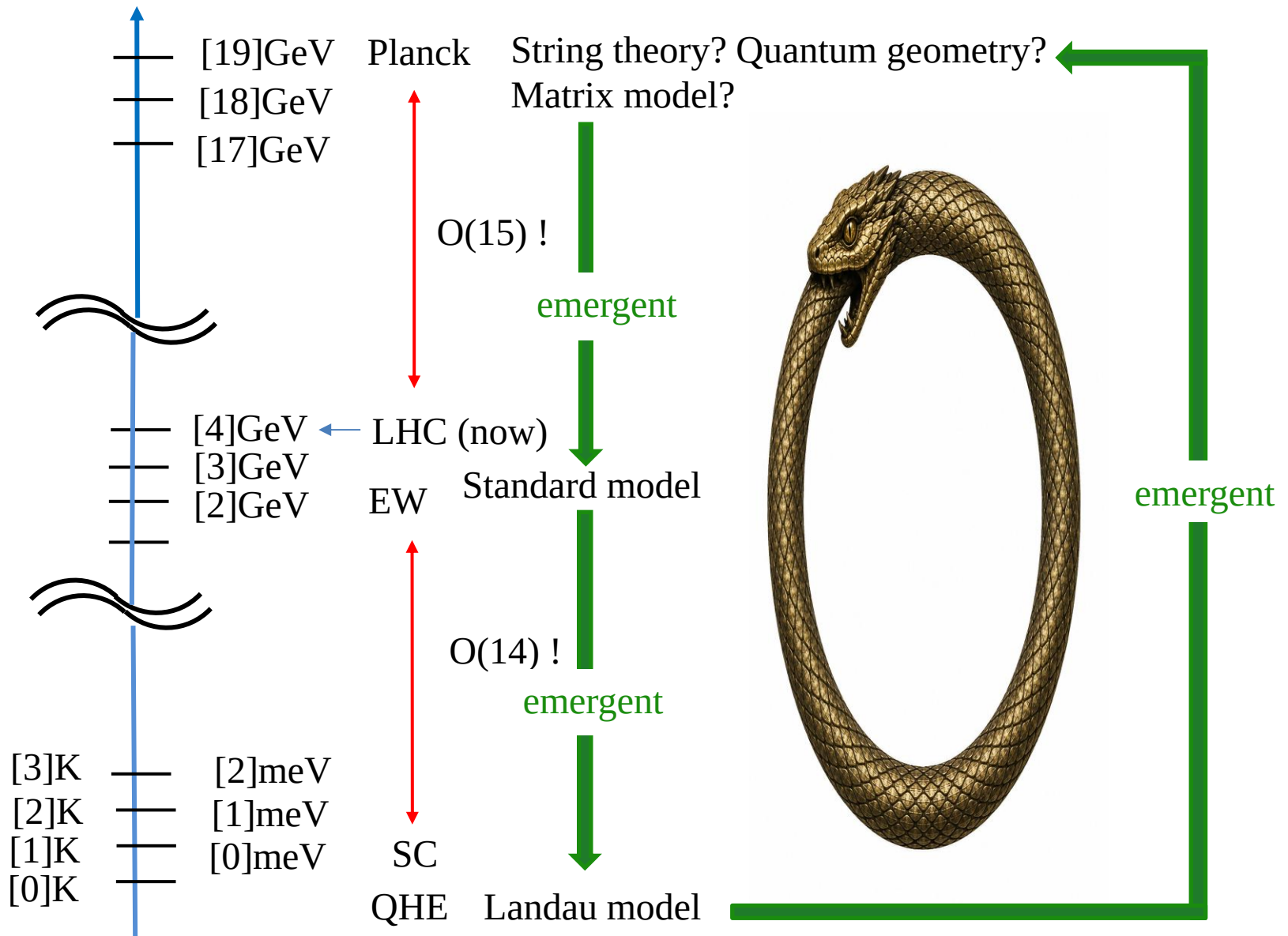
その説明になるかもしれないことを、**ポーア**が述べている。

....

彼は、「自然にあてはめることのできる数学の形式は限られた個数しかないので、
まったく誤った概念を定式化する過程で、正しい形式を見つけることもあり得る」と
言った。

Emergent uroboros from emergent phenomena

45



Summary

46

Emergent Quantum Matrix Geometry

Quantum States $\rightarrow$ Matrix Geometry

(Representation $\rightarrow$ Geometry)

$\Rightarrow$ *NC ver. ``It from bit''*

Howe duality (mathematical structure behind/beyond Landau models)

Irreps of $G \leftrightarrow$ Irreps of G'

Matrix Geometry $\leftrightarrow$ Matrix Geometry'

$\Rightarrow$ *Study on the relations among QMGs rather than each QMG*

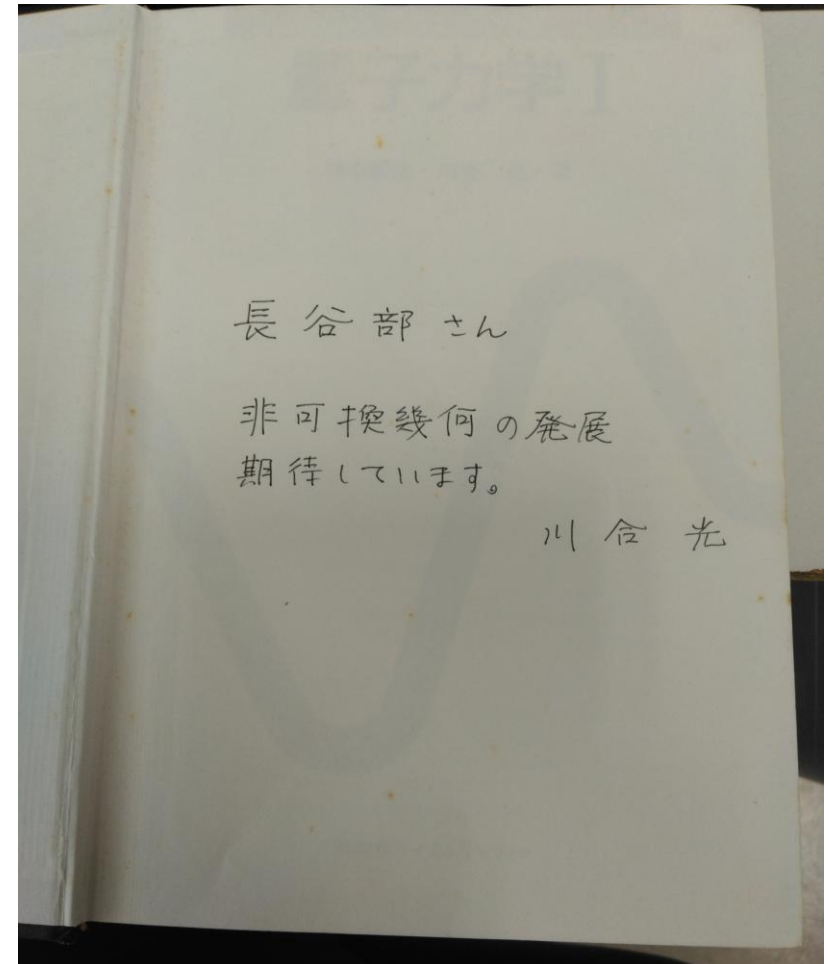
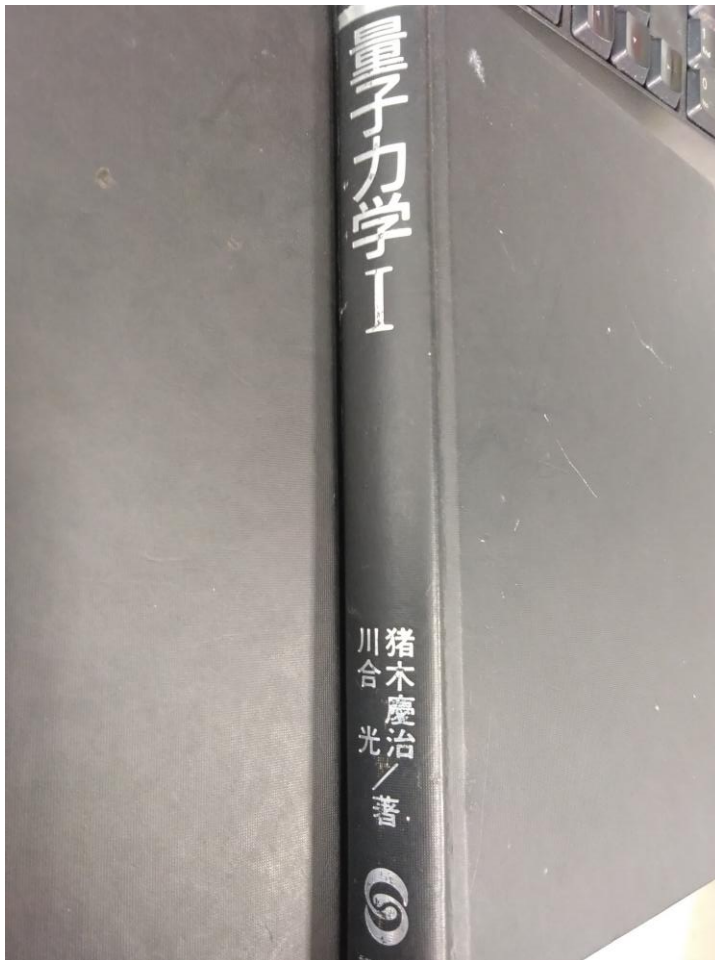
Relation to Matrix model geometry?

Internal space $\simeq$ External space

momentum A_μ *space-time coordinates*

おまけ

47



東北大の大学院の頃に(2000年くらい?), 集中講義に来られた川合先生に頂いたサイン